

Malthus Meets Romer: Population Growth Under Information Frictions

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Abstract

Households disagree about whether larger populations enhance productivity (Romerians) or diminish it (Malthusians). I embed this disagreement into a Barro–Becker fertility model with global games. Wrong beliefs generate distinct pathologies: in Romerian worlds, Malthusians produce prolonged stagnation through an *informational poverty trap*; in Malthusian worlds, Romerians produce boom-bust cycles. The framework offers two readings of the pre-industrial era: a true Malthusian regime broken by technology, or an always-Romerian world that finally escaped self-reinforcing pessimism. On the policy side, government disclosures function as Delphic forward guidance for fertility decisions. Optimal communication is truthful under dynastic welfare and shaded toward optimism under utilitarian criteria. Survey evidence shows that respondents of the Malthusian type have lower fertility.

JEL Classification: D83, J13, O40, C72

Keywords: Fertility, heterogeneous beliefs, scale effects, global games, informational traps, endogenous population welfare

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1 Introduction

How many children should a family have? The answer turns on what one believes about the economic consequences of population size. A parent who agrees with [Romer \(1990\)](#) that more people generate more ideas and faster growth may view an additional child as a contribution to future prosperity. A parent who fears, with [Malthus \(1798\)](#), that more people strain finite resources may view the same child as hastening collective impoverishment. Under intergenerational altruism, these beliefs translate directly into fertility choices, and the resulting population dynamics feed back into productivity outcomes that either reinforce or challenge them. This paper studies the consequences of that feedback loop when households hold heterogeneous beliefs about the population–productivity relationship.

I formalize the idea in an overlapping generations model in which households make fertility decisions under uncertainty about how population affects productivity growth. Households observe the same aggregate data but interpret it through different priors. Romerians believe that larger populations accelerate innovation and growth, while Malthusians believe that larger populations depress living standards. Beliefs are transmitted across generations through a combination of cultural inheritance and Bayesian updating from observed outcomes.¹ Following [Becker \(1960\)](#), children are costly in time rather than in goods: a parent devotes a fraction of time endowment to child-rearing and supplies the remainder to the labor market. Productivity itself evolves stochastically with a drift that depends on population, so today’s fertility shapes tomorrow’s productivity through the size of the next generation. The productivity-growth function is left deliberately flexible: a leading log-linear case yields closed-form expressions for calibration, but the qualitative results extend to any specification in which the Romerian and Malthusian models cross at a common reference population. These ingredients combine to yield a tractable framework in which beliefs, fertility, and growth co-evolve.

The model rests on a single behavioral premise: beliefs about population’s economic consequences shape fertility. Before drawing out theoretical implications, I establish this premise empirically. Using data from the General Social Survey (GSS), I document robust negative correlations between Malthusian beliefs (measured by agreement with the statement that Earth cannot continue to support population growth) and both ideal and actual fertility. Individuals with Malthusian beliefs report ideal fertility about 0.27 children lower than non-Malthusians and have 0.29 fewer children, gaps of roughly 11 and 16 percent of the respective sample means. The relationships persist after controlling for age, education, sex, and survey year, and they show up across demographic groups. The effect on actual fertility is at least as large as the effect on stated prefer-

¹Throughout, I use “belief” as shorthand for an individual’s structural worldview, formally modeled as a type (Romerian or Malthusian); the object that updates via Bayes’ rule is an aggregate social credence governing intergenerational transmission. [Section 2](#) makes this distinction precise.

ences, which suggests that beliefs translate into consequential behavior rather than cheap talk. The implied magnitudes are quantitatively large: a back-of-the-envelope calculation indicates that the gap between an all-Romerian and an all-Malthusian population would be roughly 0.29 children, more than half of the 0.5-child gap between current U.S. total fertility and replacement—suggesting that belief heterogeneity is a first-order force on aggregate fertility, even before accounting for the dynamic amplification mechanisms the model formalizes.²

With this empirical foundation in place, I turn to the theoretical analysis. The main theoretical contribution is to show that the dominant belief interacts with the underlying technology to generate qualitatively different pathologies depending on which worldview matches the truth. The asymmetry arises from a feature of Bayesian learning about scale effects: the informativeness of productivity signals depends on population size. When population sits near the reference level $\bar{N}$ at which the rival models predict the same growth, productivity outcomes reveal almost nothing about whether scale effects are positive or negative. Learning therefore requires population to deviate substantially from this uninformative region, but such deviations require precisely the optimistic beliefs that pessimism suppresses. This interaction between beliefs, fertility, and the informativeness of data is the engine driving the paper’s main results.

Consider first a Romerian world, where scale effects are truly positive. Here, Malthusian minorities do more than slow convergence to efficiency; they can trap the economy in prolonged stagnation. Pessimistic beliefs suppress fertility, keeping population near $\bar{N}$. A population near $\bar{N}$ generates uninformative signals about the population–productivity relationship, and uninformative signals fail to correct pessimistic beliefs. The trap is self-reinforcing: the economy remains stuck because population never deviates enough from $\bar{N}$ to reveal that Malthusianism is wrong. I call this the *informational poverty trap*. Correct beliefs eventually prevail almost surely, but the expected escape time can exceed any relevant historical horizon.

Now consider the mirror case of a Malthusian world, where scale effects are truly negative. Here the dynamics differ starkly. Demographic selection always favors optimists, who have more children regardless of whether their optimism is warranted. Bayesian learning, by contrast, favors pessimists when population growth leads to poor outcomes. The two forces pull the Romerian share ϕ_t in opposite directions, and the balance between them itself depends on the state. The paper’s second main theoretical result formalizes this opposition: in any Malthusian world, the expected drift in ϕ_t cannot be uniformly signed, because the learning channel vanishes at the crossing population (where selection dominates) and grows with the divergence of the signal distributions as population moves away from it. Monotone convergence is therefore ruled out, and the system inherits the

²The use of survey-based belief proxies follows the methodology of [Howe et al. \(2015\)](#) as adapted in [Bernstein et al. \(2019\)](#), [Baldauf et al. \(2020\)](#), and [Bakkensen and Barrage \(2022\)](#); limitations of the correlational approach are discussed in Section 3.

potential for non-monotone dynamics, including the boom-bust cycles familiar from pre-industrial demography. During expansions, optimists outbreed pessimists and population grows. Eventually the economy hits constraints, outcomes deteriorate, learning corrects beliefs toward Malthusianism, and fertility falls. Recovery begins, and the pattern can repeat.

Taken together, the two cases reveal a sharp duality: in Romerian worlds, Malthusians cause stagnation; in Malthusian worlds, Romerians generate non-monotone dynamics, including cycles. Wrong beliefs are always costly, but the nature of the cost depends on the direction of the error. The strategic structure underlying these dynamics partitions the state space into regions with distinct equilibrium properties, with multiple equilibria possible when Romerian shares are high enough that strategic complementarity dominates; the analysis uses a global-games refinement à la [Frankel et al. \(2003\)](#) to select a risk-dominant equilibrium where needed.³

The duality also offers a unified lens for interpreting demographic history. The conventional reading is that the pre-industrial world was genuinely Malthusian and that the cycles documented by economic historians ([Clark, 2007](#); [Voigtländer and Voth, 2013](#)) reflect the boom-bust dynamics of a true Malthusian regime, with the Industrial Revolution marking a technological regime change. The alternative reading, made possible by the informational poverty trap, holds that the world was always Romerian ([Kremer, 1993](#)) but stuck in a trap where population never deviated enough from the reference to escape pessimism—so that the Industrial Revolution was an *informational escape* rather than a technological one. The two readings carry distinct implications for the modern fertility decline, and Section 6 weighs the evidence between them.

The theoretical results naturally raise a policy question. If wrong beliefs can trap economies in stagnation or generate damaging cycles, can public communication help? Answering this question requires confronting what “help” even means when the population itself is endogenous to policy, since standard welfare criteria become problematic when the set of agents depends on the policy chosen ([Goloso et al., 2007](#)). I analyze optimal announcement policy under three welfare criteria—dynastic, total utilitarian, and average utilitarian—that span a substantively distinct normative spectrum and yield substantively different prescriptions. A key finding is that public information is most valuable precisely when the planner knows least: in the informational poverty trap, a credible announcement can break self-reinforcing pessimism and push population into the informative region, yet the same uninformative signals that trap households also limit what the planner can learn. The standard caution of [Morris and Shin \(2002\)](#) that public information in coordination games can amplify errors is muted here by the heterogeneous strategic interaction, in which Romerian complementarity and Malthusian substitutability provide a natural stabilizer.

³Whether the multiplicity region is empirically active depends on the strength of scale effects. Under Jones-style semi-endogenous growth calibrations ([Jones, 1995](#); [Bloom et al., 2020](#)), the region is empty; under Kremer-style or Galor–Weil-style calibrations ([Kremer, 1993](#); [Galor, 2011](#)), it is occupied at high ϕ .

This perspective suggests reframing the policy itself. Conventional family policy operates through fiscal levers—subsidized childcare, baby bonuses, parental-leave entitlements, and child tax credits—that change the price or affordability of children directly. The communication policy I analyze operates instead by shifting households’ beliefs about the long-run consequences of population, leaving the price of children unchanged. The closest analogue in macroeconomics is central-bank forward guidance and the broader practice of unconventional policy through communication (Eggertsson and Woodford, 2003; Woodford, 2005; Blinder et al., 2008; Campbell et al., 2012): a policy that works through expectations rather than through direct instrument adjustment. By analogy, I label the planner’s announcement an *unconventional family policy*. A policy contribution of the paper is to develop the welfare economics of such a policy when population is endogenous and beliefs are heterogeneous, complementing the conventional-instruments analysis surveyed in Doepke et al. (2023).

Related Literature

This paper connects several literatures: endogenous fertility, heterogeneous beliefs, global games, cultural transmission, and recent work treating prior heterogeneity as a primitive of long-run economic environments.

Endogenous fertility and population scale effects. The foundational work includes Becker (1960), who introduced the time-cost-of-children formulation I adopt, and Barro and Becker (1989), who embedded fertility choice in a dynastic optimization framework. de la Croix and Doepke (2003) pin down the time cost of one child at approximately fifteen percent of a parent’s time endowment, which I use to calibrate the curvature parameter. Subsequent work enriches this framework with human capital (Galor and Weil, 2000), mortality transitions (Kalemli-Ozcan, 2003), and intergenerational transfers (Lee, 2003); the unified growth theory of Galor (2011) synthesizes these elements. On scale effects, Romer (1990), Kremer (1993), Jones (1995, 2022), and Bloom et al. (2020) provide alternative frameworks. Ashraf and Galor (2013) finds positive long-horizon scale effects, while Voigtländer and Voth (2013) documents pre-industrial Malthusian dynamics. My contribution introduces belief heterogeneity as a driver of fertility differentials and population dynamics.

Heterogeneous beliefs. Angeletos and La’O (2013); Angeletos et al. (2018), Bordalo et al. (2018), and Malmendier and Nagel (2011) examine dispersed information and expectation formation in macroeconomics; Simsek (2013), Geanakoplos (2010), and Scheinkman and Xiong (2003) study disagreement in finance. A more recent line of work treats heterogeneous priors over slow-moving structural parameters as a primitive: Bakkensen et al. (2025) embed climate-belief disagreement

into mortgage origination, and [Gans \(2026\)](#) formalizes a strategic entry game in which prior heterogeneity is primitive, building on [Agrawal et al. \(2025\)](#) and [Felin and Zenger \(2009, 2017\)](#). My paper applies this approach to fertility.

Global games and cultural transmission. The global games literature ([Carlsson and Van Damme, 1993](#); [Morris and Shin, 1998, 2003](#); [Frankel et al., 2003](#)) provides tools for analyzing games with strategic complementarities under incomplete information. I adapt these techniques to a setting that mixes complementarities (among Romerians) and substitutabilities (among Malthusians), with the existence step relying on [Athey \(2001\)](#)'s single-crossing argument for monotone equilibria in supermodular games with continuous actions. On cultural transmission, I draw on [Cavalli-Sforza and Feldman \(1981\)](#), [Bisin and Verdier \(2001, 2011\)](#), and [Doepke and Zilibotti \(2019\)](#).

Public information and welfare with endogenous population. [Morris and Shin \(2002\)](#) shows that public information can be harmful in coordination games by crowding out private information. [Golosov et al. \(2007\)](#) provide a systematic analysis of efficiency concepts when fertility is endogenous, showing that standard welfare theorems fail under total or average utilitarianism. I build on these insights. The poverty-trap literature ([Azariadis and Drazen, 1990](#); [Banerjee and Newman, 1993](#)) studies traps from technological nonconvexities or credit constraints; the informational poverty trap I identify arises instead from endogenous uninformativeness of signals when the economy remains in a particular region of the state space.

Unconventional policy and central-bank communication. The paper's policy framework is conceptually analogous to the literature on central-bank communication and forward guidance ([Eggertsson and Woodford, 2003](#); [Woodford, 2005](#); [Blinder et al., 2008](#); [Campbell et al., 2012](#)), which studies how policy can influence outcomes by shaping expectations rather than by adjusting instruments directly. Family-policy analysis has by contrast focused almost exclusively on conventional fiscal levers such as childcare subsidies, baby bonuses, parental leave, and child tax credits ([Doepke et al., 2023](#)). To the best of my knowledge, this paper is the first to draw the analogy explicitly and to propose government information revelation as an unconventional family policy, intended to complement rather than replace conventional instruments.

The paper proceeds as follows. Section 2 presents the model. Section 3 provides empirical evidence. Section 4 analyzes strategic interaction and equilibrium selection. Section 5 examines long-run dynamics. Section 6 discusses demographic history. Section 7 analyzes optimal communication policy. Section 8 concludes. All proofs are in the Appendix.

2 Model

I develop a discrete-time overlapping generations model with dynastic altruism. The economy is populated by a continuum of households who make fertility decisions and differ in their beliefs about the structural relationship between population and productivity growth.

2.1 Environment

Time is discrete and indexed by $t = 0, 1, 2, \dots$. Each period, a generation of adults makes economic decisions, and their children become adults in the following period.

Time Endowment and Cost of Children. Each adult is endowed with one unit of time. An adult who chooses fertility n_{it} spends a fraction $\psi(n_{it}) \in [0, 1)$ of that time on child-rearing and supplies the remainder $1 - \psi(n_{it})$ to the labor market. The implicit price of an additional child is the forgone labor income. This time-cost specification, originally from [Becker \(1960\)](#), makes the resource constraint multiplicative rather than additive, which plays a central analytical role by making the household value function tractable to a first-order log-linear approximation.

Production. A single final good is produced using effective labor:

$$Y_t = A_t L_t, \quad L_t = N_t \cdot \int_0^1 (1 - \psi(n_{it})) di, \quad (1)$$

where A_t is total factor productivity, N_t is adult population, and L_t is aggregate effective labor supply.⁴ The competitive wage equals A_t , so a type- j household with fertility $n_{j,t}$ earns $A_t(1 - \psi(n_{j,t}))$. With no other income or expenditure, consumption is

$$c_{j,t} = A_t(1 - \psi(n_{j,t})). \quad (2)$$

TFP Dynamics: General Specification. I adopt a general specification of productivity dynamics that isolates which structural properties drive the paper's results. Log-productivity evolves as

$$\ln A_{t+1} = \ln A_t + g(N_t; \theta^*) + \eta_{t+1}, \quad \eta_{t+1} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad (3)$$

where $\theta^* \in \Theta$ is the scale-effect parameter, indexing the household's hypothesis about the population–productivity relationship, and $g(\cdot; \theta) : \mathbb{R}_+ \rightarrow \mathbb{R}$ is a productivity-growth function indexed by θ . The hypothesis space is $\Theta = \{\theta_R, \theta_M\}$, where Romerians entertain θ_R and Malthusians entertain θ_M ; the truth is $\theta^* \in \Theta$. The structural noise η_{t+1} is i.i.d. across periods.

⁴I interpret A_t broadly as “effective” TFP that incorporates any potential dilution effects from fixed factors.

The paper's main qualitative results require only four properties of g .

Assumption 1 (Structural Properties of g). *The growth function $g(\cdot; \theta)$ satisfies:*

- (C1) **Common crossing point.** *There exists $\bar{N} > 0$ such that $g(\bar{N}; \theta_R) = g(\bar{N}; \theta_M) \equiv \bar{g}$.*
- (C2) **Continuity at the crossing point.** *$N \mapsto g(N; \theta)$ is continuous, so the difference $g(N; \theta_R) - g(N; \theta_M)$ is continuous and vanishes as $N \rightarrow \bar{N}$.*
- (C3) **Separation away from the crossing point.** *For N outside any neighborhood of $\bar{N}$, $g(N; \theta_R) \neq g(N; \theta_M)$, and the difference grows in magnitude with $|\ln(N/\bar{N})|$ in the sense that, for any $M > 0$, there exists N with $|g(N; \theta_R) - g(N; \theta_M)| > M$.*
- (C4) **Sign orientation.** *$g(\cdot; \theta)$ is increasing in N when $\theta > 0$ (Romerian belief) and decreasing in N when $\theta < 0$ (Malthusian belief), at least in a neighborhood of $\bar{N}$; equivalently, $g(\cdot; \theta_R)$ single-crosses $g(\cdot; \theta_M)$ from below at $\bar{N}$.*

Properties (C1)–(C2) generate the informational degeneracy at $\bar{N}$: when population sits at the crossing point, the two beliefs predict identical signal distributions, so observations reveal nothing about θ^* . Property (C3) ensures that learning eventually becomes effective once population deviates substantially from $\bar{N}$. Property (C4) gives the Romerian-versus-Malthusian interpretation its economic content, ensuring that “optimism” (positive scale effect) corresponds to the belief that drives demographic expansion.

Leading Tractable Case: Log-Linear Specification. Throughout most of the paper I work with the log-linear special case

$$\ln A_{t+1} = \ln A_t + \bar{g} + \theta^* \ln(N_t/\bar{N}) + \eta_{t+1}, \quad (4)$$

which corresponds to $g(N; \theta) = \bar{g} + \theta \ln(N/\bar{N})$. This functional form satisfies Assumption 1 with $\bar{g} \geq 0$ the common deterministic component of growth at the reference, $\bar{N} > 0$ an empirically anchored reference level, and $\theta_R > 0 > \theta_M$. The log-linear form yields closed-form expressions throughout and underlies the calibrated quantitative results, but the qualitative theorems hold for any g satisfying Assumption 1. I write $g(N_t; \theta) \equiv \bar{g} + \theta \ln(N_t/\bar{N})$ for the leading case and reserve general g for results that require only Assumption 1.

A more general but still parsimonious family that also satisfies Assumption 1 is the separable class

$$g(N; \theta) = \bar{g} + \theta h(N/\bar{N}), \quad (5)$$

where $h : \mathbb{R}_+ \rightarrow \mathbb{R}$ is strictly increasing, continuous, with $h(1) = 0$ and $|h(x)| \rightarrow \infty$ as $x \rightarrow 0$ or $x \rightarrow \infty$. The leading case $h(x) = \ln x$ is one member of this family, and $h(x) = x - 1$ (a

Kremer-style linear scale effect) is another. Under (5) the KL formula (14) below generalizes by replacing $\ln(N_t/\bar{N})$ with $h(N_t/\bar{N})$, and all closed-form rates become expressions in h . The fully general specification (3), in which θ need not enter separably, also delivers the qualitative results but does not in general yield closed-form rates.

Interpretation of $\bar{N}$. When $N_t = \bar{N}$, the deterministic component of growth equals $\bar{g}$ regardless of θ^* . Observations near $N_t = \bar{N}$ are therefore uninformative about the scale-effect parameter, even though they remain noisy through η_{t+1} . This feature is central to the informational poverty trap.⁵

Households and Beliefs. The economy contains a unit mass of dynastic households indexed by $i \in [0, 1]$. Each household holds a belief θ_i about θ^* .

Assumption 2 (Belief Types). *A fraction $\phi_t \in [0, 1]$ of households are Romerians with belief $\theta_R > 0$; the remaining $1 - \phi_t$ are Malthusians with belief $\theta_M < 0$.*

A “Romerian world” has $\theta^* = \theta_R$, and a “Malthusian world” has $\theta^* = \theta_M$. Households are characterized by their type $j \in \{R, M\}$ and act on their type’s worldview as if certain: a Romerian household optimizes under $\theta = \theta_R$, and a Malthusian household optimizes under $\theta = \theta_M$. Bayesian updating in this model operates at the population level rather than at the level of individual decision-makers. We track an aggregate credence $\pi_t \in [0, 1]$ — interpretable as a public, social, or culturally shared posterior over $\{\theta_R, \theta_M\}$ — which is updated via Bayes’ rule from observed productivity outcomes and which governs the rate at which new generations are recruited into the Romerian type. The distribution ϕ_t is common knowledge; individual types are private information.⁶

Time-Cost Function.

⁵The parameter $\theta < 0$ here represents dynamic congestion or environmental drag rather than classical Malthusian fixed-factor dilution operating on the level of output. Capturing the latter would require an additional production margin (a fixed factor), which is straightforward but distinct from the channel modeled here.

⁶Although the formal apparatus uses Bayes’ rule, the cognitive interpretation in this paper is restricted to the population level. Individuals act as types under certainty; the object π_t is an aggregate social credence, and (12) is best read as a tractable functional form for socialization dynamics that is consistent with — though does not require — fully Bayesian individuals. Models of social learning in which non-Bayesian individuals generate Bayesian-like aggregate dynamics (DeGroot averaging, replicator dynamics with noise) provide micro-foundations for this interpretation. An alternative approach would allow for model ambiguity or ambiguity-averse (robust-control) behavior in the spirit of Hansen and Sargent (2001, 2008), in which agents hedge against worst-case parameter realizations rather than forming posterior means. Such a formulation would shift attention from parameter learning to model uncertainty and could make fertility choices more conservative under pessimistic scenarios. I adopt the Bayesian framework to focus on how disagreement and endogenous signal informativeness interact over time, abstracting from additional layers of ambiguity aversion.

Assumption 3 (Time Cost of Children). *The time-cost function $\psi : \mathbb{R}_+ \rightarrow [0, 1)$ satisfies (i) $\psi(0) = 0$; (ii) $\psi'(n) > 0$ and $\psi''(n) \geq 0$; (iii) $\psi(n) < 1$ for finite n with $\lim_{n \rightarrow \infty} \psi(n) = 1$; and (iv) the marginal time cost $\psi'(n)/(1 - \psi(n))$ is strictly increasing with $\lim_{n \rightarrow \infty} \psi'(n)/(1 - \psi(n)) = \infty$.*

For tractability, I use the exponential–quadratic specification

$$\psi(n) = 1 - e^{-\frac{\gamma}{2}n^2}, \quad \gamma > 0, \quad (6)$$

which yields $\psi'(n)/(1 - \psi(n)) = \gamma n$. This identity exactly mirrors the linear marginal-cost structure of a goods-cost specification while keeping consumption bounded between zero and A_t . Following [de la Croix and Doepke \(2003\)](#), the time cost of one child is approximately fifteen percent of a parent’s time endowment, which under (6) pins down $\gamma = -2 \ln(0.85) \approx 0.325$.

Preferences. Households have dynastic preferences:

$$U_{it} = u(c_{it}) + \beta n_{it} \mathbb{E}_{it} [V_{i,t+1}], \quad (7)$$

where $u(c) = \ln c$ and $\beta \in (0, 1)$ is the dynastic discount factor. Throughout, I assume equilibrium fertility satisfies $\beta n_j^* < 1$. I set $\beta = 0.30$, interpreted as a structural per-generation altruism parameter.⁷ Substituting $c_{it} = A_t(1 - \psi(n_{it}))$ gives

$$\ln c_{it} = \ln A_t + \ln(1 - \psi(n_{it})). \quad (8)$$

The first term depends only on aggregate productivity, and the second only on own fertility through the time cost.

Population Dynamics.

$$N_{t+1} = \bar{n}_t \cdot N_t, \quad \bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}, \quad (9)$$

where $n_{R,t}$ and $n_{M,t}$ denote Romerian and Malthusian fertility.

2.2 Belief Formation and Intergenerational Transmission

Cultural Transmission.

⁷This value sits at the lower end of the range used in the quantitative Barro–Becker literature ([Manuelli and Seshadri, 2009](#); [Lee and Seshadri, 2019](#); [Daruich, 2018](#)). It is conservative for the present analysis, since a higher altruism parameter would amplify the strategic complementarity slope d_R in (20) and enlarge the multiplicity region.

Assumption 4 (Cultural Transmission). *A child born to a type- j household adopts type j with probability $\lambda \in [0, 1]$ and forms beliefs through learning with probability $1 - \lambda$.*

Bayesian Learning. The aggregate credence π_t is updated by Bayes' rule from observed log-productivity growth. This update can be interpreted as a reduced-form description of social learning — through education, public discourse, and cultural transmission of inferences from aggregate data — rather than as a within-head computation performed by any individual household. Observed log-productivity growth is

$$s_t \equiv \ln(A_{t+1}/A_t) = g(N_t; \theta^*) + \eta_{t+1}. \quad (10)$$

Given s_t and N_t , the likelihood under belief θ is

$$\ell(s_t | \theta, N_t) = \frac{1}{\sqrt{2\pi}\sigma_\eta} \exp\left(-\frac{(s_t - g(N_t; \theta))^2}{2\sigma_\eta^2}\right), \quad (11)$$

and Bayes' rule gives

$$\pi_{t+1} = \frac{\pi_t \cdot \ell(s_t | \theta_R, N_t)}{\pi_t \cdot \ell(s_t | \theta_R, N_t) + (1 - \pi_t) \cdot \ell(s_t | \theta_M, N_t)}. \quad (12)$$

Children who learn adopt Romerian beliefs with probability π_{t+1} and Malthusian beliefs with probability $1 - \pi_{t+1}$.

Signal Informativeness. The Kullback–Leibler divergence between the signal distributions under the two beliefs is

$$\mathcal{I}_t \equiv \text{KL}(\ell(\cdot | \theta_R, N_t) \parallel \ell(\cdot | \theta_M, N_t)) = \frac{[g(N_t; \theta_R) - g(N_t; \theta_M)]^2}{2\sigma_\eta^2}. \quad (13)$$

By (C1), $\mathcal{I}_t = 0$ at $N_t = \bar{N}$, so the signal is completely uninformative about θ^* . By (C2), $\mathcal{I}_t$ is continuous in N_t , so it vanishes as $N_t \rightarrow \bar{N}$. By (C3), $\mathcal{I}_t$ becomes arbitrarily large as population deviates substantially from $\bar{N}$. Under the log-linear leading case, (13) reduces to

$$\mathcal{I}_t = \frac{(\theta_R - \theta_M)^2 [\ln(N_t/\bar{N})]^2}{2\sigma_\eta^2}. \quad (14)$$

More generally, under the separable specification (5), $\mathcal{I}_t = (\theta_R - \theta_M)^2 h(N_t/\bar{N})^2 / (2\sigma_\eta^2)$. The rate at which $\mathcal{I}_t$ vanishes near $\bar{N}$ governs the quantitative rate at which the informational trap binds: if $g(N; \theta_R) - g(N; \theta_M) = O(|\ln(N/\bar{N})|^k)$ near $\bar{N}$, then $\mathcal{I}_t = O(|\ln(N/\bar{N})|^{2k})$. The log-linear case

yields $k = 1$, and higher k intensifies the trap.

Evolution of Belief Distribution. Combining cultural transmission and learning, the Romerian share evolves as

$$\phi_{t+1} = \lambda \phi_t^{birth} + (1 - \lambda) \pi_{t+1}, \quad (15)$$

where $\phi_t^{birth} = \phi_t n_{R,t} / \bar{n}_t$ is the fraction of children born to Romerians.

Independence of Types. One modeling choice deserves explicit mention. In the model above, a household's type θ_i and other households' types are drawn independently from the population distribution induced by (ϕ_t, π_t) . Knowing one's own θ_i does not lead a household to update its belief about other households' types. This assumption matches a world in which type heterogeneity comes from idiosyncratic sources: different parental theories, different tacit knowledge, different analogical reasoning, and different tastes for risk-taking in the construction of a mental model. A household that happens to be very optimistic learns nothing about other households' optimism from its own conviction, because its conviction is generated by an idiosyncratic internal theory rather than a common external signal.⁸ The cultural-transmission parameter λ introduces a within-dynasty correlation that does not affect the cross-section independence assumption. The public-validation extension in Section 7 introduces a different limited form of correlation, in which a public announcement shifts the aggregate social credence π_t by the same amount for all households while leaving the residual idiosyncratic component independent.

2.3 Equilibrium

Definition 1 (Equilibrium). *An equilibrium consists of fertility policies $n_R(A_t, N_t, \phi_t, \pi_t)$, $n_M(A_t, N_t, \phi_t, \pi_t)$; aggregate fertility $\bar{n}(A_t, N_t, \phi_t, \pi_t)$; and laws of motion for $(A_{t+1}, N_{t+1}, \phi_{t+1}, \pi_{t+1})$ given by (3), (9), (15), and (12), such that (a) each household's fertility maximizes (7) subject to (2) given the household's type and the aggregate fertility function; (b) expectations about aggregate fertility are consistent with the equilibrium aggregate; and (c) actual laws of motion use the true θ^* and realized η_{t+1} .*

The Bellman below treats a type- j parent's continuation value as the value function of a type- j child, even though the child may switch with probability $1 - \lambda$.⁹

⁸This is the natural counterpart of the theory-based view of strategic disagreement (Felin and Zenger, 2009, 2017). If type heterogeneity were instead generated by a common source such as a publicly observable disclosure, then a household's own optimistic type assignment would rationally shift its belief about others' types upward, and individual types would be positively correlated conditional on (ϕ_t, π_t) . The clean closed-form best-response slopes derived in Section 4 would no longer obtain in the same form.

⁹An exact formulation would replace $\mathbb{E}_j[V_j]$ by $\lambda \mathbb{E}_j[V_j] + (1 - \lambda) \mathbb{E}_j[V_{learn}]$, where V_{learn} is a type-mixture

2.4 Household Optimization

The Bellman for a type- j household reads

$$V_j(A, N, \phi, \pi) = \max_{n \geq 0} \{ \ln A + \ln(1 - \psi(n)) + \beta n \cdot \mathbb{E}_j[V_j(A', N', \phi', \pi')] \}. \quad (16)$$

Under the type interpretation, $\mathbb{E}_j[\cdot]$ denotes expectation under the type- j law of motion $g(\cdot; \theta_j)$, with no within-household uncertainty over θ . The aggregate credence π_t does not enter the household FOC directly; it enters the model only through the recruitment dynamics (15).

I work with a first-order log-linear approximation around a balanced-growth reference at $N = \bar{N}$. I conjecture

$$V_j(A, N, \phi, \pi) \approx v_j(\phi, \pi) + \xi_{A,j} \ln A + \xi_{N,j} \ln(N/\bar{N}), \quad (17)$$

with $(\xi_{A,j}, \xi_{N,j})$ and $v_j(\phi, \pi)$ determined by coefficient matching. Under the general specification (3), the approximation expands $g(N; \theta)$ to first order around $\bar{N}$:

$$g(N; \theta) \approx \bar{g} + g_N(\bar{N}; \theta) \bar{N} \ln(N/\bar{N}) + o(\ln(N/\bar{N})).$$

The technology side is therefore approximately linear in $\ln(N/\bar{N})$ with local slope $g_N(\bar{N}; \theta) \bar{N}$. Under the leading log-linear case this slope is exactly θ , and the approximation is exact in $\ln(N/\bar{N})$. The first-order condition for fertility is

$$\frac{\psi'(n_j)}{1 - \psi(n_j)} = \beta \mathbb{E}_j[V_j(A', N', \phi', \pi')], \quad (18)$$

which under (6) simplifies to $\gamma n_j = \beta \mathbb{E}_j[V_j]$.

Lemma 1 (Equilibrium Fertility, Local Coefficients). *Under Assumptions 1–4 and the local log-linear conjecture (17), the value function coefficients evaluated at the balanced-growth reference are*

$$\xi_{A,j} = \frac{1}{1 - \beta n_j^*}, \quad \xi_{N,j} = \frac{\beta n_j^* \tilde{\theta}_j}{(1 - \beta n_j^*)^2}, \quad (19)$$

where $\tilde{\theta}_j \equiv g_N(\bar{N}; \theta_j) \bar{N}$ is the local elasticity of growth with respect to population under belief θ_j ; under the log-linear leading case $\tilde{\theta}_j = \theta_j$. Equilibrium fertility satisfies a fixed-point equation in n_j^* and can be written as $n_j^* = n^* h(\theta_j; \bar{n})$ with h increasing in θ . Moreover, $n_R^* > n_M^*$ for all equilibrium values of $\bar{n}$ under (C4).

The proof is in Appendix A. Households who believe population enhances productivity (Rome-continuation. The two formulations coincide when λ is close to 1 or when V_R and V_M are close in the relevant region; both conditions hold near the steady states studied below.

rians) perceive a higher return to children and choose higher fertility. This testable prediction motivates the empirical analysis in the next section.

3 Empirical Evidence: Beliefs and Fertility

A central prediction of the model is that Malthusian beliefs reduce fertility. This section examines empirical support using survey data.

3.1 Data

I use the General Social Survey (GSS), a nationally representative survey of U.S. adults. The key belief variable comes from the question: “The earth cannot continue to support population growth at its present rate.” Respondents indicate agreement on a five-point scale, and I classify those who agree or strongly agree as holding *Malthusian beliefs*. This question was asked in the 2000 and 2010 waves.¹⁰ I examine two fertility measures: stated preferences (“ideal number of children”) and realized fertility (actual number of children). The sample contains 1,822 observations.

3.2 Descriptive Patterns

Table 1 presents summary statistics. Mean ideal fertility is 2.50 (s.d. 0.87) and mean actual fertility is 1.81 (s.d. 1.64). Approximately 55 percent of respondents hold Malthusian beliefs.

Table 1. Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Ideal number of children	1,822	2.50	0.87	0	7
Actual number of children	1,822	1.81	1.64	0	8
Malthusian belief (binary)	1,822	0.55	0.50	0	1
Malthusian belief (1–5 scale)	1,822	3.44	1.03	1	5
Age	1,822	46.8	17.3	18	89
Years of education	1,822	13.3	2.96	0	20
Female	1,822	0.56	0.50	0	1

Notes: Data from GSS, 2000 and 2010 waves. Malthusian belief (binary) equals one if respondent agrees or strongly agrees that Earth cannot continue to support population growth.

Figure 1 displays mean ideal and actual fertility by Malthusian belief intensity. There is a clear monotonic negative relationship between belief strength and both fertility measures. Respondents

¹⁰The use of survey-measured belief proxies has a substantial track record in environmental and climate finance (Howe et al., 2015; Bernstein et al., 2019; Baldauf et al., 2020; Bakkensen and Barrage, 2022; Bakkensen et al., 2025), with validation evidence that aggregated belief measures correlate strongly with door-to-door individual surveys.

who strongly disagree report an average ideal of 2.98 children and have 1.76 children, while those who strongly agree report 2.24 and 1.52.

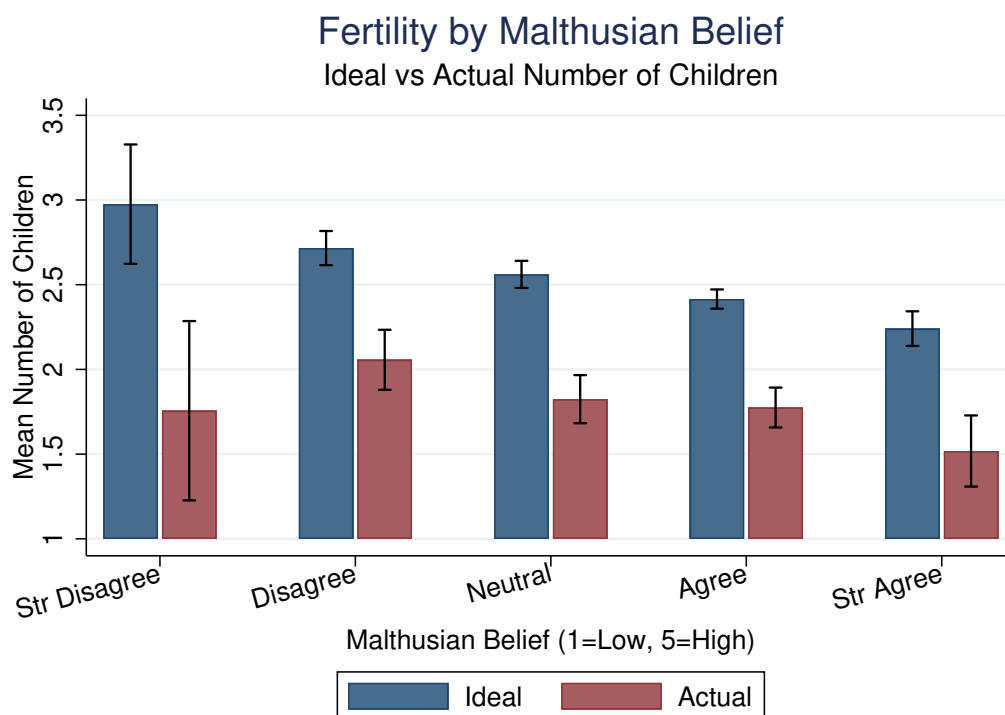


Figure 1. Fertility by Malthusian Belief Intensity

Notes: Data from GSS, 2000 and 2010 waves (N = 1,822). Bars show mean ideal (navy) and actual (maroon) number of children for each level of the Malthusian belief scale. Error bars indicate 95% confidence intervals.

A natural concern is that this relationship is driven by education. Figure 2 shows mean ideal fertility by education group and belief type. Within every education category, Malthusians report lower ideal fertility than non-Malthusians, with gaps ranging from 0.20 to 0.45 children.

3.3 Regression Analysis

Table 2 presents linear regressions of both ideal and actual fertility on Malthusian beliefs with demographic controls.¹¹

Malthusians report ideal fertility 0.27 children lower than non-Malthusians (column 1), about 11 percent of the sample mean. The coefficient on actual fertility is -0.29 (column 3), implying Malthusians have approximately 16 percent fewer children. The coefficient on actual fertility

¹¹All specifications include both age and age squared, so the coefficient on age in the table reflects the linear component only. The marginal effect of age combines linear and quadratic components and is therefore age-specific.

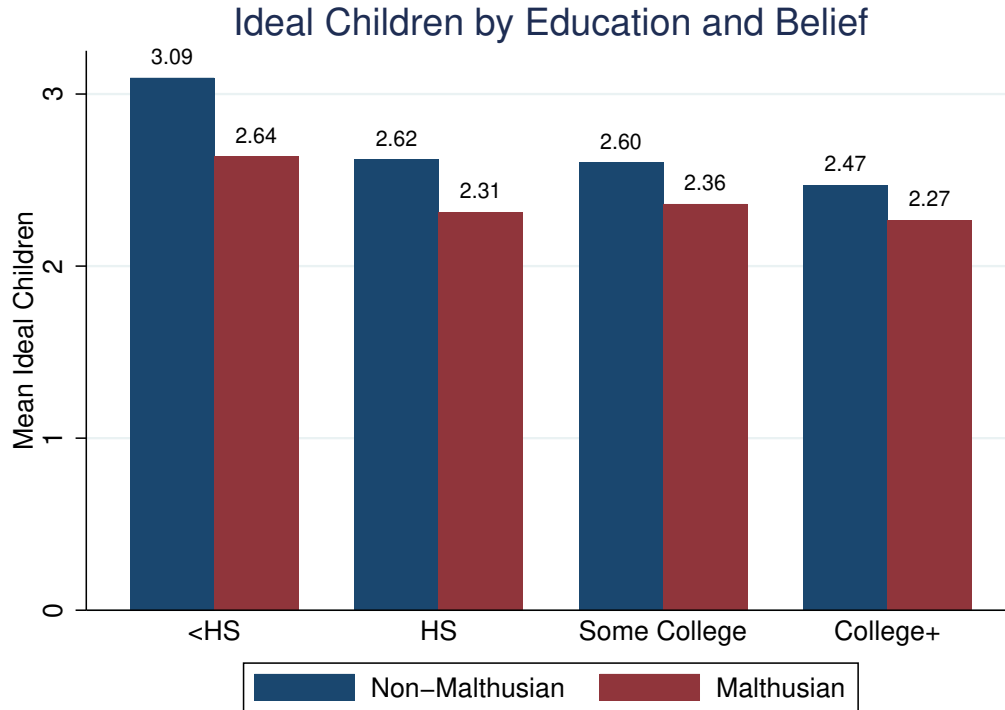


Figure 2. Ideal Number of Children by Education and Malthusian Belief

Notes: Bars show mean ideal number of children for non-Malthusians (navy) and Malthusians (maroon) within each education category.

is larger in absolute value than the coefficient on ideal fertility, suggesting beliefs translate into behavior with at least as much force as they shape stated preferences.

3.4 Completed Fertility and Heterogeneity

A concern with actual fertility is that younger respondents may not have completed childbearing. Table 3 restricts the sample to women aged 45+ and men aged 50+. Among those with completed fertility, Malthusians have 0.26 fewer children than non-Malthusians, similar to the full-sample estimate.

Table 4 examines heterogeneity by education. The belief–fertility relationship is present across education groups for both outcome measures. For ideal fertility, the coefficient is somewhat larger for less educated respondents; for actual fertility, the pattern reverses, with the coefficient larger among the more educated. The descriptive pattern is consistent with the interpretation that Malthusian beliefs affect fertility preferences across the education distribution but translate into behavior more strongly among the more educated, perhaps because they face fewer constraints on achieving their fertility targets.

Table 2. Effect of Malthusian Beliefs on Fertility

	Ideal Fertility		Actual Fertility	
	(1)	(2)	(3)	(4)
Malthusian	−0.268*** (0.041)		−0.292*** (0.066)	
Malthusian (continuous)		−0.153*** (0.021)		−0.146*** (0.033)
Age	−0.014** (0.006)	−0.015** (0.006)	0.112*** (0.010)	0.111*** (0.010)
Female	0.015 (0.040)	0.026 (0.040)	0.091 (0.068)	0.104 (0.068)
Education	−0.047*** (0.007)	−0.046*** (0.007)	−0.144*** (0.013)	−0.143*** (0.013)
Year FE	Yes	Yes	Yes	Yes
Observations	1,822	1,822	1,822	1,822
R^2	0.060	0.069	0.262	0.262

Notes: All models include age squared and year fixed effects. Robust standard errors in parentheses. The reported age coefficient is the linear term only. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3. Actual Fertility by Completed Fertility Status

	Full Sample (1)	Completed Fertility (2)	Not Yet Completed (3)
Malthusian	−0.292*** (0.066)	−0.257** (0.109)	−0.288*** (0.079)
Observations	1,822	848	974
R^2	0.262	0.112	0.266

Notes: Dependent variable is actual number of children. All models control for age, age squared, sex, education, and year. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.5 Interpretation, Aggregate Implications, and Limitations

The empirical findings are consistent with the model's central prediction: individuals who believe population growth is unsustainable both desire and have fewer children. The relationships are economically meaningful and statistically robust. Two features deserve emphasis. First, the relationship holds for actual fertility, as well as for stated preferences. Second, the coefficient on actual fertility is at least as large as the coefficient on ideal fertility.

Beyond statistical significance, the micro estimates imply effects that are economically mean-

Table 4. Heterogeneity in the Belief–Fertility Relationship

	Ideal Fertility		Actual Fertility	
	HS or Less	Some College+	HS or Less	Some College+
Malthusian	−0.318*** (0.069)	−0.235*** (0.049)	−0.209* (0.115)	−0.392*** (0.081)
Observations	821	1,001	821	1,001
R^2	0.045	0.029	0.163	0.240

Notes: Dependent variable is ideal (columns 1–2) or actual (columns 3–4) number of children. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

ingful at the aggregate level. Aggregate fertility in the model is $\bar{n} = \phi n_R + (1 - \phi) n_M$, so the marginal aggregate effect of a shift in the Romerian share is $\partial \bar{n} / \partial \phi = n_R - n_M$. Taking the actual-fertility coefficient from Table 2 as a measure of this gap gives $n_R - n_M \approx 0.29$. A counterfactual population in which all respondents held non-Malthusian beliefs would have mean fertility approximately $1.81 + 0.55 \times 0.29 \approx 1.97$, while a fully Malthusian population would have approximately $1.81 - 0.45 \times 0.29 \approx 1.68$. The implied range, roughly 1.68 to 1.97, brackets most of the gap between current U.S. total fertility (about 1.6) and replacement (2.1) and spans a substantial fraction of the cross-country variation in completed fertility among rich economies. Equivalently, a ten-percentage-point shift in the Malthusian share of the population moves aggregate fertility by approximately 0.03 children—roughly one tenth of the gap between current U.S. fertility and replacement. These calculations are deliberately conservative: they treat the micro coefficient as a static partial equilibrium response and ignore the dynamic amplification mechanisms that the model formalizes, including cultural transmission, Bayesian feedback, and demographic selection in favor of optimists, all of which compound the static effect over generations.

Several limitations are worth flagging. The evidence remains correlational and could reflect reverse causality or omitted-variable bias; the cross-sectional coefficient may also understate the aggregate effect if beliefs and behavior co-move at the social-network or community level, and may overstate it if part of the cross-sectional gap reflects unobserved heterogeneity correlated with both beliefs and fertility. The cross-sectional nature of the data precludes tracking how belief changes affect subsequent fertility decisions. The belief measure also captures concern about global population sustainability rather than specifically the scale-effect parameter θ .¹² Despite these

¹²A natural template for stronger identification would combine survey-based belief proxies with high-resolution geographic variation in exposure to belief-relevant shocks (extreme weather events, salient environmental news, or pronatalist policy changes), or use repeated cross-sections to track within-county changes in beliefs against subsequent fertility. The GSS data alone do not support such designs, and I view causal identification as a priority for follow-up empirical work.

limitations, the evidence—together with the back-of-the-envelope calculation above—provides suggestive support that the belief channel is quantitatively, and not only qualitatively, relevant for aggregate fertility dynamics.

4 Strategic Interaction and Equilibrium Selection

This section analyzes strategic interaction in fertility choices and partitions the equilibrium space into three economically distinct regions. Three results organize the rest of the paper. First, Romerians exhibit strategic complementarity while Malthusians exhibit strategic substitutability, so the slope of the aggregate best response is a continuous function of the Romerian share that switches sign at an interior threshold. Second, two thresholds ϕ^* and ϕ^{**} partition the share space into a Malthusian-dominated region of unconditional uniqueness, a contraction region of conditional uniqueness, and a Romerian-dominated multiplicity region whose presence depends on the strength of scale effects. Third, in the multiplicity region a global-games refinement à la [Frankel et al. \(2003\)](#) selects a risk-dominant equilibrium.

4.1 The Fertility Game

Consider fertility choice as a game among households. Each household i chooses n_i to maximize expected utility, and aggregate fertility is $\bar{n} = \int n_i di$. The household's payoff depends on $(n_i, \bar{n})$ through a single channel: higher $\bar{n}$ raises $N' = \bar{n}N$, which through (3) affects A'' and subsequent productivity. Let $U_j(n_j, \bar{n})$ denote the expected utility of a type- j household when aggregate fertility is $\bar{n}$. Type j exhibits *strategic complementarity* if $\partial^2 U_j / \partial n_j \partial \bar{n} > 0$ and *strategic substitutability* if the cross-partial is negative.

4.2 Best Responses and the Sign of Strategic Interaction

Let $n_j(\bar{n}^e)$ denote the fertility choice of type j when expected aggregate fertility is $\bar{n}^e$. Rational expectations require $\bar{n}^e = \bar{n}$. Under the local log-linear approximation, the marginal effect of expected aggregate fertility on the continuation value is $\partial \mathbb{E}_j[V_j] / \partial \bar{n}^e = \xi_{N,j} / \bar{n}^e$. From (19), the sign of $\xi_{N,j}$ matches the sign of $\tilde{\theta}_j = g_N(\bar{N}; \theta_j) \bar{N}$, which by (C4) coincides with the sign of θ_j , so that $\xi_{N,M} < 0$ for Malthusians and $\xi_{N,R} > 0$ for Romerians. Since the household FOC has a strictly increasing left-hand side in n_j , the sign of $dn_j^* / d\bar{n}^e$ coincides with the sign of $\xi_{N,j}$.

Proposition 1 (Strategic Interaction). *Under Assumptions 1–4: (i) Romerians exhibit strategic complementarity, $dn_R^* / d\bar{n}^e > 0$; (ii) Malthusians exhibit strategic substitutability, $dn_M^* / d\bar{n}^e < 0$.*

The closed-form local slopes are

$$d_j \equiv \left. \frac{dn_j^*}{d\bar{n}^e} \right|_{ref} = \frac{\beta^2 n_j^* \tilde{\theta}_j}{\gamma \bar{n} (1 - \beta n_j^*)^2}, \quad (20)$$

where $\tilde{\theta}_j = g_N(\bar{N}; \theta_j) \bar{N}$; under the log-linear leading case $\tilde{\theta}_j = \theta_j$.

The proof is in Appendix A. A Romerian who expects higher aggregate fertility also expects higher future population, which under Romerian beliefs implies faster productivity growth and a higher continuation value of each child. A Malthusian who expects higher aggregate fertility expects slower productivity growth, which lowers the continuation value and reduces fertility. The strategic structure of the game depends on g only through (C4), namely the sign of $\partial g / \partial N$ evaluated at $\bar{N}$.

4.3 Three Equilibrium Regions

Define the aggregate best-response slope

$$\rho(\phi) \equiv \phi d_R + (1 - \phi) d_M = \phi d_R - (1 - \phi) |d_M|, \quad (21)$$

with $d_R > 0 > d_M$. $\rho(\phi)$ crosses zero at

$$\phi^* = \frac{|d_M|}{d_R + |d_M|}, \quad (22)$$

and crosses one at

$$\phi^{**} = \frac{1 + |d_M|}{d_R + |d_M|}, \quad (23)$$

provided the right-hand side is below one, which holds iff $d_R > 1$. The two thresholds carve the unit interval into three economically distinct regions.

Theorem 1 (Equilibrium Existence and Uniqueness by Region). *Suppose Assumptions 1–4 hold and Proposition 1 applies at the relevant reference. Then:*

- (i) **Malthusian-dominated region**, $\phi < \phi^*$. *The aggregate best response $BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1 - \phi) n_M^*(\bar{n}^e)$ is strictly decreasing in a neighborhood of any candidate fixed point. The equilibrium is locally unique unconditionally on the magnitude of $\rho(\phi)$.*
- (ii) **Contraction region**, $\phi^* \leq \phi \leq \phi^{**}$ (or all $\phi \geq \phi^*$ when $d_R \leq 1$). *$0 \leq \rho(\phi) < 1$, the contraction-mapping argument applies, and the equilibrium is locally unique.*

(iii) **Multiplicity region**, $\phi > \phi^{**}$ (only when $d_R > 1$). $\rho(\phi) > 1$ and BR can intersect the 45-degree line transversally at multiple points, generating an odd number of locally ordered equilibria of which the extreme two are stable.

The proof is in Appendix A. The result depends on g only through the signs and local magnitudes of $\tilde{\theta}_R$ and $\tilde{\theta}_M$, so the specific functional form is immaterial. In the Malthusian-dominated region, optimist coordination is too thin to generate the strategic complementarity needed for multiple stable points. In the contraction region, complementarity is bounded, so the standard contraction argument delivers uniqueness. Only in the Romerian-dominated region with sufficiently strong scale effects does complementarity dominate enough to generate multiplicity.

When does the multiplicity region exist? Theorem 1(iii) requires $d_R > 1$, which substituting (20) reads

$$\tilde{\theta}_R > \tilde{\theta}_R^{crit} \equiv \frac{\gamma \bar{n} (1 - \beta n_R^*)^2}{\beta^2 n_R^*}. \quad (24)$$

At the calibration $(\beta, \gamma) = (0.30, 0.325)$ with $n_R^* = 1.2$ and $\bar{n} \approx 1$, $\tilde{\theta}_R^{crit} \approx 1.23$. Under the log-linear leading case this corresponds to $\theta_R > 1.23$. Under standard semi-endogenous-growth calibrations the multiplicity region is empty, while under Kremer–Galor–Weil calibrations it is occupied at high ϕ . Both cases are interesting.

4.4 Stability and Global-Games Selection

Within the multiplicity region, local stability provides a first-pass refinement: an equilibrium is stable iff $|\rho(\phi)| < 1$ at the slopes evaluated locally at that equilibrium. With $\rho(\phi) > 1$ on average, the typical configuration features two stable equilibria flanking an unstable middle equilibrium. Stability eliminates the middle equilibrium but leaves the two stable selections to be coordinated upon.

To break this remaining indeterminacy, I adopt the global-games refinement of Frankel et al. (2003). The key technical step is a reduction. Because Malthusians have a strictly downward-sloping best response and a unique fixed point given any aggregate fertility level, their best response can be substituted out. The reduced game among Romerians is supermodular and falls within the scope of Frankel et al. (2003).¹³

Theorem 2 (Global-Games Selection in the Multiplicity Region). *Consider the multiplicity region of Theorem 1(iii). Suppose households observe private signals $\phi_i = \phi + \zeta_i$ with $\zeta_i \sim \mathcal{N}(0, \sigma_\zeta^2)$ and*

¹³The construction of monotone pure best responses uses the single-crossing approach of Athey (2001), which delivers a non-empty, compact, convex-valued, closed-graph best-response correspondence to which Kakutani's theorem can be applied.

the prior on ϕ is diffuse. As $\sigma_\zeta \rightarrow 0$, iterated deletion of strictly dominated strategies in the reduced Romerian subgame selects a unique threshold equilibrium with cutoff $\phi^{rd}(\theta_R, \theta_M, \beta, \gamma) \in (\phi^{**}, 1)$ at which Romerians coordinate on the high-fertility selection above the cutoff and on the low-fertility selection below. The selected equilibrium is risk-dominant in the sense of [Frankel et al. \(2003\)](#).

The proof is in Appendix [A](#). The result restores a unique equilibrium selection across the entire ϕ -space.

Why selection matters for policy. Outside the multiplicity region, an announcement plays a single role: it shifts households' subjective posteriors over θ , and through them the unique equilibrium fertility. Inside the multiplicity region, an announcement plays a second role as well, since by shifting the prior over ϕ it can move the economy across the cutoff ϕ^{rd} and switch the selection. The selection-shifting role is most relevant when the prior ϕ is near ϕ^{rd} and largely absent when the prior is far from it. Section [7](#) treats both roles.

4.5 Map of Regions and Empirical Relevance

Table [5](#) summarizes the partition. The trap region of Section [5](#) typically features ϕ below the critical share at which aggregate fertility just reaches replacement, which is well below ϕ^{**} for any standard calibration. The trap therefore sits inside the contraction (or Malthusian-dominated) region, and the policy intuition for the trap is unaffected by multiplicity. The interior steady state of the Malthusian world also sits in the contraction region for standard calibrations.

Table 5. Equilibrium Regions and Their Economic Content

Region	Condition	$\rho(\phi)$	Equilibrium	Where it applies
Malthusian-dominated	$\phi < \phi^*$	< 0	Unique unconditionally	Trap; pre-industrial era
Contraction	$\phi^* \leq \phi \leq \phi^{**}$	$[0, 1)$	Unique via contraction	Most calibrations
Multiplicity	$\phi > \phi^{**}$	> 1	Multiple, selection by GG	Strong scale effects

Notes: Multiplicity region is non-empty iff $\tilde{\theta}_R > \tilde{\theta}_R^{crit}$, with $\tilde{\theta}_R^{crit} \approx 1.23$ at the baseline calibration $(\beta, \gamma) = (0.30, 0.325)$ and $n_R^* = 1.2$. GG denotes the global-games selection of Theorem [2](#).

The stabilizing role of Malthusians. Because Malthusian best responses slope downward, the Malthusian fraction $1 - \phi$ acts as a continuous shifter that pushes $\rho(\phi)$ down. Compared to a population of only Romerians, the heterogeneous mix delivers a larger contraction region and a smaller multiplicity region. Pessimists, paradoxically, stabilize the equilibrium structure of the economy.

5 Long-Run Dynamics and the Role of Wrong Beliefs

I now analyze the long-run dynamics of beliefs and population. Wrong beliefs generate qualitatively different pathologies depending on which worldview is correct, and the asymmetry arises from the interaction between demographic selection, which always favors optimists, and Bayesian learning, which favors correct beliefs but only when signals are informative. Throughout this section I state explicitly which properties of g each result requires.

5.1 Two Forces: Demographic Selection and Learning

The Romerian share ϕ_t evolves according to (15), driven by two forces.

Demographic selection. Since $n_R > n_M$, $\phi_t^{birth} > \phi_t$ for any interior ϕ_t . Demographic selection pushes the cultural-transmission part of ϕ_{t+1} upward regardless of which belief is correct.

Bayesian learning. The direction of π_{t+1} depends on the true θ^* . The strength of learning depends on population through the signal informativeness $\mathcal{I}_t$ in (13). When $N_t \approx \bar{N}$, (C1) and (C2) imply $\mathcal{I}_t \approx 0$ and learning is weak regardless of the truth.

In Romerian worlds the two forces align; in Malthusian worlds they conflict. The next two subsections develop each case in turn.

5.2 Dynamics in a Romerian World: Convergence and the Trap

Theorem 3 (Convergence in Romerian Worlds). *Suppose $\theta^* = \theta_R$ (Romerian world) and Assumption 1 holds; in particular, by (C3), $\mathcal{I}(N)$ is bounded away from zero whenever N is outside any neighborhood of $\bar{N}$. Suppose further that the population trajectory remains within the region where (3) is well-defined. Starting from any interior initial condition $(\phi_0, N_0, A_0, \pi_0)$: (i) $\pi_t \rightarrow 1$ almost surely; (ii) $\phi_t \rightarrow 1$ almost surely; (iii) aggregate fertility converges to n_R^* ; (iv) $\sup_t \ln(N_t/\bar{N}) = +\infty$ almost surely.*

The proof is in Appendix A. Both forces favor Romerians when the world is truly Romerian. The result uses (C1)–(C4): (C1)–(C2) define the uninformative region; (C3) ensures that $\sum_t \mathcal{I}_t = \infty$ a.s., the key step in the martingale-convergence argument; and (C4) ensures $n_R > n_M$ and hence positive demographic-selection drift.¹⁴

¹⁴The result that $\sup_t \ln(N_t/\bar{N}) = +\infty$ presumes the absence of endogenous bounds on population. The framework can be extended to deliver bounded population through a hard carrying capacity or endogenous congestion. Under either extension, $\phi_t \rightarrow 1$ almost surely, but population converges to a stochastic neighborhood of the carrying capacity. The informational poverty trap is unaffected because it operates at populations close to $\bar{N}$.

Theorem 3 guarantees almost-sure convergence but is silent on the rate. The next analysis shows that convergence can be arbitrarily slow.

The Uninformative Region.

Definition 2 (Uninformative Region). *For $\delta > 0$, the δ -uninformative region is $\{N_t : |\ln(N_t/\bar{N})| < \delta\}$. By (C1)–(C2), the per-period information bound on this region satisfies $\mathcal{I}_t \leq \bar{\mathcal{I}}(\delta)$, where $\bar{\mathcal{I}}(\delta) \rightarrow 0$ as $\delta \rightarrow 0$. Under the log-linear leading case, $\bar{\mathcal{I}}(\delta) = (\theta_R - \theta_M)^2 \delta^2 / (2\sigma_\eta^2)$.*

When δ is small, signals reveal almost nothing about θ^* . The rate at which $\bar{\mathcal{I}}(\delta) \rightarrow 0$ depends on the local behavior of $g(N; \theta_R) - g(N; \theta_M)$ near $\bar{N}$: if this difference is of order $|\ln(N/\bar{N})|^k$, then $\bar{\mathcal{I}}(\delta) = O(\delta^{2k})$. The log-linear case has $k = 1$, while a model with quadratic local divergence has $k = 2$ and yields slower escape.

Definition 3 (Trap Region). *The informational poverty trap region $\mathcal{T}(\delta, \zeta)$ is the set of states (ϕ, N, π) with: (i) $|\ln(N/\bar{N})| < \delta$; (ii) $\bar{n}(\phi) \in [1, 1 + \zeta]$ for small $\zeta > 0$; and (iii) $\pi \leq \phi$, so posterior pessimism aligns with prevalent Malthusianism.*

The third condition ensures the trap is closed under the joint dynamics of (ϕ_t, N_t, π_t) . Once $\pi_t \leq \phi_t$, this inequality propagates inductively (see proof of Proposition 2).

Assumption 5 (Fertility Straddles Replacement). *Malthusian fertility is below replacement and Romerian fertility above: $n_M < 1 < n_R$.*

Under Assumption 5, there is a critical Romerian share $\phi^{trap} = (1 - n_M)/(n_R - n_M)$ at which $\bar{n} = 1$. For ϕ slightly above ϕ^{trap} , aggregate fertility slightly exceeds 1 and population grows slowly. ϕ^{trap} lies within the Malthusian-dominated or contraction regions for standard calibrations.

Proposition 2 (Slow Learning in the Trap). *Consider a Romerian world ($\theta^* = \theta_R > 0$) under Assumptions 1 and 5. Suppose $(\phi_0, N_0, \pi_0) \in \mathcal{T}(\delta, \zeta)$ for small $\delta, \zeta > 0$. For target $\bar{\phi} > \phi_0$, define the escape time $T^{esc} = \inf\{t : \phi_t > \bar{\phi}\}$. Then, as long as the trajectory remains in $\mathcal{T}(\delta, \zeta)$:*

- (i) $|\mathbb{E}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t]| \leq C_1 \bar{\mathcal{I}}(\delta)$ with $C_1 = 1/4$;
- (ii) $\phi_{t+1}^{birth} - \phi_t \leq C_2(n_R - n_M)(1 - \phi_t)$ with $C_2 = 1/n_M$;
- (iii) $\mathbb{E}[T^{esc}] \geq (\bar{\phi} - \phi_0)/D$, where $D = \lambda C_2(n_R - n_M) + (1 - \lambda)C_1 \bar{\mathcal{I}}(\delta)$.

The proof is in Appendix A. The result uses only (C1)–(C2). As $\delta \rightarrow 0$, the learning contribution vanishes by (C2) and escape relies on demographic selection, which is itself slow when $n_R - n_M$ or λ is small. Corollary 3 shows that the expected escape time diverges at a rate determined by the local behavior of $g(N; \theta_R) - g(N; \theta_M)$ near $\bar{N}$.

Proposition 7 characterizes the boundary of the trap. The critical population beyond which learning dominates demographic selection is, under the log-linear leading case,

$$N^{crit}(\phi)/\bar{N} = \exp\left(\sqrt{\frac{2\sigma_\eta^2 \cdot \lambda C_2(n_R - n_M)(1-\phi)}{(1-\lambda)C_1(\theta_R - \theta_M)^2}}\right).$$

Under the general separable specification (5), the corresponding critical population is the solution to $h(N^{crit}/\bar{N})^2 = 2\sigma_\eta^2 \lambda C_2(n_R - n_M)(1-\phi)/[(1-\lambda)C_1(\theta_R - \theta_M)^2]$. High noise σ_η^2 raises N^{crit} , making the trap larger; strong cultural transmission λ also raises N^{crit} ; and large belief dispersion $(\theta_R - \theta_M)^2$ lowers N^{crit} by making signals more diagnostic.

In a Romerian world, then, Malthusian minorities cause prolonged stagnation through a self-reinforcing mechanism: pessimism suppresses fertility, low fertility keeps population near $\bar{N}$, and (C1)–(C2) imply that uninformative signals fail to correct pessimism. Escape eventually occurs but can take arbitrarily long.¹⁵

5.3 Dynamics in a Malthusian World: Opposing Forces and Non-Monotone Outcomes

When $\theta^* = \theta_M$, demographic selection and learning pull ϕ_t in opposite directions.

Proposition 3 (Opposing Forces in Malthusian Worlds). *Suppose $\theta^* = \theta_M < 0$ and Assumption 1 holds. At any interior state with $\phi_t \in (0, 1)$ and $\pi_t \in (0, 1)$:*

- (i) *Demographic-selection drift on births is strictly positive and bounded. The contribution of differential fertility to the share of Romerian-parented children satisfies*

$$0 < \phi_t^{birth} - \phi_t = \frac{\phi_t(1-\phi_t)(n_R - n_M)}{\bar{n}_t} \leq \frac{n_R - n_M}{4n_M},$$

independently of N_t .

- (ii) *Bayesian learning drift on the posterior is strictly negative under truth and scales with signal informativeness:*

$$\mathbb{E}^{\theta_M}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t] = -\pi_t(1-\pi_t)\mathcal{I}_t + o(\mathcal{I}_t), \quad \mathcal{I}_t = \frac{[g(N_t; \theta_R) - g(N_t; \theta_M)]^2}{2\sigma_\eta^2}.$$

By (C1), $\mathcal{I}_t = 0$ at $N_t = \bar{N}$; by (C3), $\mathcal{I}_t$ becomes arbitrarily large for population sufficiently far from $\bar{N}$.

¹⁵This mechanism differs sharply from settings where delay is informative: in some dynamic-learning models (e.g., Gans, 2026), silence carries information and high variance only postpones rather than prevents action. Here, the analog of silence (population at the reference $\bar{N}$) is by construction uninformative about θ^* , so trap durations can be unboundedly large rather than merely lengthy.

(iii) The expected drift in ϕ_t admits a sign change as a function of population deviation. Using (15),

$$\mathbb{E}^{\theta_M}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t] = \underbrace{\lambda(\phi_t^{birth} - \phi_t)}_{\geq 0, \text{ selection}} + \underbrace{(1 - \lambda)(\pi_t - \phi_t)}_{\text{cultural regression toward } \pi_t} + \underbrace{(1 - \lambda)\mathbb{E}^{\theta_M}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t]}_{\leq 0, \text{ learning}}.$$

At $N_t = \bar{N}$ the learning term vanishes by (C1); for population sufficiently far from $\bar{N}$, (C3) implies the learning term dominates and the drift is strictly negative whenever $\pi_t \in (0, 1)$. The deterministic skeleton admits no monotone law of motion in ϕ_t .

The proof, in Appendix A, is a direct Bayes log-odds calculation combined with a bounded-selection-drift comparison. It uses (C1), (C3), and (C4); the specific functional form of g is not needed. Under the log-linear leading case, the threshold takes the form

$$\bar{L}(\phi_t, \pi_t, \lambda)^2 = \frac{2\sigma_\eta^2 \cdot [\lambda(\phi_t^{birth} - \phi_t) + (1 - \lambda) \max\{\pi_t - \phi_t, 0\}]}{(1 - \lambda) \pi_t (1 - \pi_t) (\theta_R - \theta_M)^2},$$

expressed in units of $\ln(N_t/\bar{N})^2$. Under the general specification, the threshold is in units of $[g(N_t; \theta_R) - g(N_t; \theta_M)]^2$, and existence of a sign-changing state follows from continuity and (C3).

Economic content. Demographic selection always pushes ϕ_t upward because optimists outbreed pessimists, while learning under truth always pushes downward because outcomes confirm the Malthusian model, and the magnitude of the learning channel scales with how far population has deviated from the reference, as measured by the divergence of the rival growth functions. The drift is therefore non-negative near $\bar{N}$ and strictly negative once population has grown enough that the learning channel overtakes selection. Any candidate interior steady state must lie at a population strictly above the reference, where $g(N^{ss}; \theta_M) = 0$; under the log-linear leading case this gives $\ln(N^{ss}/\bar{N}) = -\bar{g}/\theta_M > 0$.

Cycle dynamics. The state-dependent reversal naturally produces the cyclical pattern familiar from economic-historical accounts of the pre-industrial era, with three identifiable phases. In the expansion phase, population is moderate, learning is weak by (C2), demographic selection dominates, and ϕ_t rises. In the crisis phase, as ϕ_t and population grow, (C3) implies signals become informative; productivity slows because $\theta^* = \theta_M$; the Malthusian model fits much better; π_{t+1} falls sharply, overwhelming demographic selection; and ϕ_{t+1} falls. In the recovery phase, fertility falls, population stabilizes or contracts, signals weaken, demographic selection regains the upper hand, and the pattern can repeat. Whether the system traces out a deterministic limit cycle, damped oscillations, or stochastic recurrence depends on curvature of the nonlinear dynamics and

shock size.

Comparative statics for the strength of cycles. When learning is very weak (small $(\theta_R - \theta_M)^2$ or large σ_η^2), $\bar{L}$ is large and the drift remains positive over a wide population range, so demographic selection dominates and ϕ_t rises persistently. When learning is very strong, $\bar{L}$ is small and the drift turns negative for modest population deviations, so Malthusian beliefs spread quickly and the system settles into a Malthusian-dominated regime. The most pronounced non-monotone dynamics arise for intermediate learning responsiveness, where neither force dominates globally.

5.4 Synthesis: The Asymmetry of Wrong Beliefs

The two preceding subsections deliver a sharp asymmetry, summarized in Table 6. The mechanism depends on the population–productivity relationship only through (C1)–(C4); the specific functional form of g governs quantitative rates but not qualitative conclusions.

Table 6. Pathologies from Wrong Beliefs

	Romerian World ($\theta^* > 0$)	Malthusian World ($\theta^* < 0$)
Wrong belief	Malthusian ($\theta_M < 0$)	Romerian ($\theta_R > 0$)
Demographic selection	Favors Romerians (correct)	Favors Romerians (incorrect)
Learning	Favors Romerians (correct)	Favors Malthusians (correct)
Forces align?	Yes	No
Pathology	Informational poverty trap	Non-monotone dynamics, including cycles
Long-run outcome	Convergence to truth	State-dependent reversal

The economic implication is that human societies are more vulnerable to excessive optimism in terms of stability, but more vulnerable to excessive pessimism in terms of stagnation. Optimists reproduce more regardless of whether their optimism is warranted, which destabilizes Malthusian worlds; pessimism keeps population near the reference and thereby suppresses learning, which traps Romerian worlds. Wrong beliefs are always costly, but the nature of the cost depends on the direction of the error.

5.5 Robustness: What Would Break the Results

The four conditions (C1)–(C4) are more than sufficient. Each plays an identifiable role, and relaxing any of them changes the conclusions in substantive ways. It is useful to spell out what would go wrong if any were dropped.

1. **Without (C1) (no common crossing point).** If $g(N; \theta_R) > g(N; \theta_M)$ for every N , no population level is uninformative. Signals always discriminate, learning is always strong, and

the informational poverty trap vanishes. The Malthusian-world opposing-forces mechanism also weakens, since selection and learning never trade off at a specific population range. Theorem 3 continues to hold (and indeed convergence is faster), while Proposition 2 and the trap intuition do not.

2. **Without (C2) (discontinuity at the crossing point).** If the divergence $\mathcal{I}(N)$ jumps at $\bar{N}$, the uninformative region has measure zero and households learn essentially instantly upon any small population deviation. Proposition 2(i) fails because $\bar{\mathcal{I}}(\delta)$ does not vanish with δ .
3. **Without (C3) (no eventual separation).** If $g(N; \theta_R) - g(N; \theta_M)$ remains bounded as $|N/\bar{N}|$ grows, the informativeness of signals is capped. Theorem 3 may fail because $\sum_t \mathcal{I}_t < \infty$ along bounded trajectories, and the Malthusian-world sign change in Proposition 3(iii) may not occur. Cycles in particular require sufficient eventual separation.
4. **Without (C4) (no sign orientation).** The correspondence between “optimism” and Romerianism loses its economic content. Demographic selection still favors high-fertility households, but those need no longer be Romerians, so the asymmetry between Romerian-world traps and Malthusian-world cycles dissolves.

A separate concern is the assumption that the noise variance σ_η^2 is constant in N . If $\sigma_\eta^2(N) \rightarrow 0$ as $N \rightarrow \bar{N}$ at the same rate as $\mathcal{I}(N)$ vanishes, the signal-to-noise ratio remains bounded away from zero at the crossing point, and the trap dissolves. The trap requires that signal noise does not vanish faster than the rival-model gap at the crossing point, a substantive but mild restriction.

A further concern is uniqueness of the crossing. If $g(N; \theta_R) = g(N; \theta_M)$ at multiple values of N , the state space contains multiple informational traps, one around each crossing. The long-run dynamics become correspondingly richer, with the economy potentially shuttling between traps. The qualitative existence of traps and cycles survives, with the global picture becoming more involved.

6 Demographic History: Two Interpretations

The model offers a unified framework for interpreting broad patterns in demographic history. It admits two fundamentally different interpretations of the pre-industrial era, with distinct implications for understanding the modern world.

6.1 Interpretation 1: A Genuinely Malthusian Past

The conventional interpretation, consistent with economic historians from Malthus himself to [Clark \(2007\)](#), holds that the pre-industrial world was genuinely Malthusian, with negative scale effects

and larger populations reducing living standards.

Under this interpretation, the cycles documented by Voigtländer and Voth (2013) and others reflect exactly the kind of non-monotone dynamics established in Proposition 3. During periods of relative prosperity, Romerian-like optimism spread through demographic selection, population grew, and eventually population deviation from the reference became large enough that the learning channel engaged. Living standards then fell, Malthusian beliefs were reinforced, and the drift in ϕ_t reversed.

The Black Death of 1347–1351 provides a particularly clear example. Europe’s population had grown substantially, straining resources. The plague killed roughly one-third of the population. Real wages rose sharply in the aftermath and remained elevated for over a century. The model interprets this as a crisis phase that both directly reduced population and reinforced Malthusian beliefs.

Under this interpretation, the Industrial Revolution marked a genuine regime change from $\theta^* < 0$ to $\theta^* > 0$. Sustained technological progress, enabled by the accumulation of scientific knowledge and supporting institutions, broke the Malthusian link.

6.2 Interpretation 2: An Always-Romerian World Trapped by Information Poverty

The informational poverty trap opens a radically different interpretation: the world may have always been Romerian, with humanity trapped for millennia by the self-reinforcing dynamics of pessimism.

Under this interpretation, positive scale effects existed throughout history, but populations were too close to the relevant reference, and productivity signals too noisy, for anyone to reliably learn this fact. Malthusian beliefs, once established, suppressed fertility and kept population near $\bar{N}$. The trap persisted because the evidence needed to refute Malthusianism could only be generated by escaping it.

The observed pre-industrial cycles, under this interpretation, were failed escape attempts from the trap rather than the inevitable boom-bust dynamics of a Malthusian world. Occasionally, some region or era experienced conditions that pushed population higher: better weather, new crops, reduced warfare. Beliefs began to update toward optimism. The gains were fragile. Any negative shock such as plague, famine, or invasion was interpreted as confirmation of Malthusian fears, beliefs snapped back, and the economy returned to the trap.

The Industrial Revolution, under this interpretation, was an *informational escape*. Some combination of factors, including the Enlightenment, improving institutions, and a sequence of positive productivity shocks, pushed European populations above the critical threshold N^{crit} . Once in the informative region, learning accelerated, evidence of positive scale effects accumulated, beliefs shifted decisively toward Romerianism, fertility rose further, and population exploded.

6.3 Distinguishing the Interpretations

Both interpretations are consistent with the broad qualitative features of demographic history, with each marshaling distinct evidence in its favor.

The case for genuine regime change. Several pieces of evidence support the view that the pre-industrial world was truly Malthusian and that the Industrial Revolution constituted a fundamental shift in the underlying technology of production. First, the empirical fingerprint of Malthusian dynamics in pre-1800 data is robust: [Clark \(2007\)](#) documents that English real wages oscillated around a stationary mean for centuries, with positive shocks (such as the Black Death studied by [Voigtländer and Voth \(2013\)](#)) raising wages temporarily before population recovery erased the gains. [Ashraf and Galor \(2013\)](#) provide complementary cross-country evidence that pre-industrial productivity differences were absorbed by population density rather than living standards, exactly as a Malthusian model predicts. Second, the timing of the takeoff coincides with concrete technological discontinuities—the steam engine, mechanized textiles, and the systematic exploitation of fossil fuels—rather than with a gradual accumulation of ideas, suggesting a qualitative break in the production technology itself ([Mokyr, 1990, 2002](#); [Allen, 2009](#)). Third, Unified Growth Theory ([Galor and Weil, 2000](#); [Galor, 2011](#)) provides an internally consistent account in which the transition is endogenous: rising returns to human capital eventually trigger a quantity–quality substitution and the demographic transition, but only after the economy has slowly escaped the Malthusian trap through millennia of latent technological accumulation. Under this interpretation, the apparent stationarity of pre-industrial living standards represents the genuine equilibrium of a fundamentally different economic regime rather than a veil over an underlying Romerian technology.

The case for informational escape. The alternative view—that scale effects in idea production were always present and that history reflects belief-driven movement between equilibria of a single underlying model—also has substantial empirical and theoretical support. The most influential evidence comes from [Kremer \(1993\)](#), who shows that over the very long run the growth rate of population is increasing in the level of population, exactly as a scale-effects model predicts. Kremer’s cross-regional test is particularly powerful: after the post-Pleistocene separation of the continents, regions with larger initial populations (Eurasia, the Americas, Australia, Tasmania) experienced faster technological progress over the subsequent millennia, isolating the scale effect from any regime-change story. This pattern is difficult to reconcile with the strict view that scale effects switched on only in 1800. Second, the speed of the post-1800 transition—a doubling of living standards within a few generations after millennia of stagnation—is more naturally consistent with a movement between equilibria than with the gradual accumulation of a new technology ([Jones, 2003](#)). Third, large pre-industrial civilizations, such as Song China, appear to have approached the

threshold for takeoff but failed to cross it, consistent with a coordination-failure interpretation in which the technology was capable of supporting growth but beliefs and institutions did not align.

Taking stock. I do not take a definitive stand. The two interpretations need not be mutually exclusive: the world may have been predominantly Malthusian in its first-order dynamics while scale effects in ideas operated as a latent second-order force, with belief coordination determining whether and when the threshold was crossed. The contribution of the model is to show that both interpretations are theoretically coherent within a unified framework and that they have sharply different implications for policy, particularly for how governments should respond to the modern fertility decline.

6.4 The Modern Fertility Decline

Fertility rates in developed countries have fallen dramatically since the 1960s, reaching levels well below replacement throughout Europe and East Asia and, more recently, in much of the developing world ([Doepke et al., 2023](#)). The framework suggests three non-exclusive interpretations of this decline.

First, the decline may reflect a revival of Malthusian beliefs. The environmental movement, concerns about overpopulation popularized in the 1960s and 1970s ([Ehrlich, 1968](#); [Meadows et al., 1972](#)), and contemporary climate anxiety can all be understood as a resurgence of the view that additional people impose net costs on existing ones. The empirical evidence in Section 3 documents that such beliefs are widespread and quantitatively associated with substantially lower fertility, consistent with related findings on how climate-related beliefs shape economic decisions ([Bakkensen and Barrage, 2022](#); [Bakkensen et al., 2025](#); [Baldauf et al., 2020](#)). Under the informational-escape interpretation, the modern fertility decline represents a partial drift back toward the Malthusian equilibrium, driven by a coordination of beliefs on a more pessimistic view of the returns to population rather than by a change in the underlying technology.

Second, the decline may reflect a genuine shift in the underlying technology. Under the regime-change interpretation, the true sign of θ^* may have recently changed from positive to negative. Several developments are consistent with this view: automation and artificial intelligence may have reduced the marginal contribution of additional human labor; ideas may be becoming harder to find, blunting the scale effect even at constant population ([Bloom et al., 2020](#)); and binding environmental constraints may impose genuine costs on population growth that were absent in earlier eras. Under this interpretation, low fertility is an efficient response to a changed economic environment rather than a coordination failure, with very different policy implications.

Third, the decline may reflect quantity–quality substitution that is conceptually orthogonal to Malthusianism but generates similar fertility outcomes. The model abstracts from human capital,

but in reality parents face the trade-off emphasized by [Becker \(1960\)](#) and [Barro and Becker \(1989\)](#): rising returns to education induce parents to invest more intensively in fewer children ([Becker and Lewis, 1973](#); [de la Croix and Doepke, 2003](#); [Manuelli and Seshadri, 2009](#); [Doepke, 2005](#); [Doepke and Zilibotti, 2019](#)). This mechanism can lower fertility even when beliefs about the social returns to population remain optimistic and when the underlying technology continues to exhibit scale effects. Empirically distinguishing the quantity–quality channel from the belief-driven channel is an important task, and the survey evidence in [Section 3](#) suggests that both are operative: respondents differ both in their beliefs about the externalities of fertility and in their human-capital investment plans, and these two margins are only weakly correlated.

These interpretations have markedly different policy implications. If low fertility reflects a belief-driven coordination failure, then public communication and expectations management— analogous to forward guidance in monetary policy ([Eggertsson and Woodford, 2003](#); [Woodford, 2005](#); [Campbell et al., 2012](#))—can in principle move the economy back toward the high-fertility equilibrium at low fiscal cost. If low fertility reflects a genuine technological shift, then such communication policies are ineffective or even harmful, and the appropriate response is to accept lower steady-state population. If low fertility reflects quantity–quality substitution, then policies should target the relative price of child quantity and quality (e.g., childcare subsidies, parental leave) rather than beliefs or aggregate population. Disentangling these mechanisms empirically is therefore a prerequisite for pronatalist policy design.

7 Policy: Welfare Criteria and the Value of Public Information

I now analyze the welfare effects of public announcements about the scale-effect parameter θ . The analysis combines an information-provision channel that operates throughout the share space and a selection-shifting channel that operates inside the multiplicity region.

7.1 Welfare Criteria with Endogenous Population

Evaluating welfare when fertility is a choice variable raises fundamental conceptual issues, because the set of people who exist depends on policy.

Dynastic Welfare. Under the dynastic criterion, the planner respects households’ subjective preferences:

$$W^{dyn} = \phi_0 V_R(A_0, N_0, \phi_0, \pi_0) + (1 - \phi_0) V_M(A_0, N_0, \phi_0, \pi_0). \quad (25)$$

This is closely related to the P-efficiency concept of [Golosov et al. \(2007\)](#).

Total Utilitarian Welfare.

$$W^{tot} = \mathbb{E}^* \left[\sum_{t=0}^{\infty} \beta^t N_t \cdot \ln c_t \right]. \quad (26)$$

For (26) to be well-defined, $\beta \mathbb{E}^* n_R^* < 1$ at the relevant equilibrium fertility. Under the time-cost specification, $\ln c_t = \ln A_t + \ln(1 - \psi(n_t))$ decomposes additively.

Average utilitarian benchmark.

$$W^{avg} = \mathbb{E}^* \left[\sum_{t=0}^{\infty} \beta^t \ln c_t \right]. \quad (27)$$

W^{avg} internalizes the per-capita scale-effect channel but omits the population-multiplier term N_t .

The dynastic and total utilitarian criteria span a substantively distinct normative spectrum. The dynastic planner functions primarily as an information provider, while the total utilitarian planner functions also as an externality corrector. The average-utilitarian benchmark sits between them.

7.2 The First-Best Allocation and Welfare Decomposition

A planner who knows θ^* and can directly choose a symmetric fertility path under the total utilitarian criterion solves

$$\max_{\{n_t\}} \mathbb{E}^* \left[\sum_{t=0}^{\infty} \beta^t N_t \cdot (\ln A_t + \ln(1 - \psi(n_t))) \right] \quad (28)$$

subject to $N_{t+1} = n_t N_t$ and (3). Conjecturing $V^P(A, N) = N \cdot w^P(A, N)$ with $w^P(A, N) \approx v^P + \xi_A^P \ln A + \xi_N^P \ln(N/\bar{N})$, the planner's FOC for fertility is

$$\frac{\psi'(n_t^{FB})}{1 - \psi(n_t^{FB})} = \beta [w^P(A', N') + \xi_N^P]. \quad (29)$$

The household's common-correct-belief FOC is structurally identical except for the absence of ξ_N^P . The wedge between planner and household optimality is exactly $\beta \xi_N^P$, with the sign of θ^* , which by (C4) coincides with the sign of $g_N(\bar{N}; \theta^*)$.

Proposition 4 (First-Best vs. Decentralized Fertility). *In a Romerian world ($\theta^* > 0$): (i) $n^{FB} > n^{CB}$, where n^{CB} is the common-correct-belief decentralized fertility; (ii) $n^{FB} > n^{CB} > n_M^*$; (iii) the welfare gap decomposes into externality, mean-bias, and dispersion components. In a Malthusian world ($\theta^* < 0$), $n^{FB} < n^{CB}$.*

The proof is in Appendix B.

Proposition 5 (Welfare Decomposition). *Under the total utilitarian criterion,*

$$W^{FB}(\theta^*) - W^{HB} = \underbrace{\Delta_1(\theta^*)}_{\text{externality}} + \underbrace{\Delta_2(\theta^*, \bar{\theta})}_{\text{mean bias}} + \underbrace{\Delta_3(\text{Var}(\theta_i))}_{\text{dispersion}}, \quad (30)$$

where $\Delta_1 \geq 0$ with equality iff $\theta^* = 0$; $\Delta_2 \geq 0$ is increasing in $(\bar{\theta} - \theta^*)^2$; and $\Delta_3 \approx (\omega\kappa^2/2)\text{Var}(\theta_i)$.¹⁶

The proof is in Appendix B. Public announcements can reduce Δ_2 and Δ_3 but cannot directly address Δ_1 without additional policy instruments. The decomposition delivers three messages. First, even with $\bar{\theta} = \theta^*$ so $\Delta_2 = 0$, the externality Δ_1 persists. Second, even with no dispersion and correct mean beliefs, the externality remains. Third, dispersion generates welfare loss even when $\bar{\theta} = \theta^*$.

7.3 The Planner's Announcement Problem: Two Channels

Consider a planner who observes a noisy estimate of the true parameter,

$$\hat{\theta} = \theta^* + \nu, \quad \nu \sim \mathcal{N}(0, \sigma_\nu^2), \quad (31)$$

and publicly releases a signal $z = \mu + \tilde{\nu}$, where $\tilde{\nu} \sim \mathcal{N}(0, \tau_A^{-1})$. The announcement operates through two channels.

(1) Information-provision channel. The announcement z enters the aggregate Bayesian updating rule (12) as an additional public signal, shifting the social posterior π_t^{post} toward the announced value. Through (15), this in turn shifts the recruitment of the next generation toward Romerian (or Malthusian) types and so changes ϕ_{t+1} . Existing households' types are unchanged by the announcement; the policy operates entirely through socialization.

(2) Selection-shifting channel. In the multiplicity region, the announcement also serves as a public signal about which equilibrium will be played. Through Theorem 2, an announcement can move the cutoff ϕ^{rd} and switch the selection between low-fertility and high-fertility stable equilibria.

The two channels reinforce each other in Romerian worlds with strong scale effects.

Assumption 6 (Announcement Credibility). *Households treat the announcement as a Gaussian signal with precision $\tau_A > 0$. When $\tau_A \rightarrow \infty$, households fully adopt the announced value; when $\tau_A \rightarrow 0$, households ignore the announcement.*

¹⁶The Taylor expansion underlying Δ_3 is taken around the social-optimum fertility n^{SP} , so the externality Δ_1 appears explicitly rather than being absorbed.

7.4 Optimal Announcement under Different Criteria

Theorem 4 (Optimal Announcement: Dynastic Welfare). *Under W^{dyn} , the optimal announcement is truthful in expectation: $\mu^{dyn} = \hat{\theta}$, and the planner prefers maximal informativeness.*

Under the dynastic criterion, the planner functions primarily as an information provider. The planner has no externality motive beyond what households themselves would weight, so distorting the announcement strictly reduces welfare.

Theorem 5 (Optimal Announcement: Total Utilitarian Welfare). *Consider $\theta^* = \theta_R > 0$. Under W^{tot} , the optimal announcement satisfies*

$$\mu^{tot} = \hat{\theta} + \chi^{tot}, \quad (32)$$

where $\chi^{tot} > 0$ is an “optimism shade” whenever the externality $\Delta_1 > 0$. The shade has the following comparative statics: (i) χ^{tot} is increasing in $|\theta^*|$; (ii) χ^{tot} is increasing in $|\ln(N_0/\bar{N})|$; (iii) χ^{tot} is decreasing in $\text{Var}(\theta_{i,0})$; (iv) χ^{tot} is increasing in announcement credibility τ_A . In a Malthusian world, the sign of χ^{tot} reverses.

The proof is in Appendix B. The intuition is that the planner equates the marginal utilitarian benefit of shifting average fertility (carrying an externality wedge of size $\beta\xi_N^P$) to the marginal cost of distorting individual fertility decisions away from their belief-conditional optimum.

The same logic, applied to a planner who omits the population-multiplier term, yields a strictly smaller shade, since the relevant wedge no longer carries the additional motive to expand population.

Corollary 1 (Optimal Announcement: Average Utilitarian Welfare). *Under W^{avg} , the optimal announcement is $\mu^{avg} = \hat{\theta} + \chi^{avg}$, where $0 < \chi^{avg} < \chi^{tot}$ in a Romerian world (with the inequalities reversing in a Malthusian world). The wedge driving χ^{avg} is the per-capita scale-effect spillover $\beta\xi_{N,avg}^P$, which is strictly smaller in absolute value than $\beta\xi_N^P$ because it omits the population-multiplier term N_t .*

Stacking the three criteria yields a clean ordering, which is itself a substantive normative result.

Corollary 2 (Ordering of Announcements). *In a Romerian world, $\mu^{dyn} = \hat{\theta} < \mu^{avg} < \mu^{tot}$, with the inequalities reversing in a Malthusian world.*

7.5 The Selection-Shifting Channel and the Planner’s Dilemma

When the economy is in or near the multiplicity region, the announcement operates additionally through the selection channel. The cutoff ϕ^{rd} is a function of expected fertility levels under each

selection, both of which respond to households' subjective θ^{post} . An optimism-shaded announcement therefore not only raises Romerian fertility on the high-fertility selection but also lowers the cutoff at which the high-fertility selection is played.

Proposition 6 (Selection-Shifting Effect). *In the multiplicity region, the optimal total utilitarian announcement satisfies $\mu^{tot,sel} = \mu^{tot} + \chi^{sel}$, where $\chi^{sel} > 0$ in a Romerian world. The magnitude of χ^{sel} is non-monotone in ϕ : it peaks when ϕ is near ϕ^{rd} and vanishes when ϕ is far from it on either side.*

The proof is in Appendix B. When the economy is firmly in the high-fertility basin, shifting the cutoff has no effect on the realized selection. When firmly in the low-fertility basin, the cutoff is far from binding. Selection-shifting matters only at the margin.

A key insight emerges from the structure of the learning problem: the planner faces the same informational constraints as households. If the economy is in the uninformative region, the historical productivity signals that inform $\hat{\theta}$ are themselves uninformative. The planner's signal precision is endogenously low precisely when the trap is most binding. Two state-contingent consequences follow, with announcement values formalized below.

In the trap region in a Romerian world, the expected value of truthful announcement under W^{tot} takes the form $\mathcal{V}^{trap} = P^{esc} \cdot \Delta W^{esc} - P^{deepen} \cdot \Delta W^{deepen}$ (Appendix B, Proposition 8), with a precision threshold above which $\mathcal{V}^{trap} > 0$. Because ΔW^{esc} is large—reflecting the welfare gain of escaping prolonged stagnation—even a moderately noisy signal can be worth announcing. By contrast, in a Malthusian world with non-monotone dynamics, the value of announcement comes from reducing belief dispersion and thereby dampening oscillations, a benefit partially offset by adjustment costs from disrupting the existing dynamics (Proposition 9). The required precision threshold is correspondingly higher.

Remark 1 (The Planner's Dilemma). *Public information is most valuable in the informational poverty trap, where coordinating beliefs could push population above N^{crit} and initiate escape. But this is precisely when the planner has the least information. The cases where intervention matters most are the cases where the planner is most uncertain about what to say.*

This creates a policy dilemma distinct from Morris and Shin (2002). There, the concern is that public information may be too influential. Here, the planner may have nothing useful to say when saying something would matter most. In the informational poverty trap, even a moderately noisy announcement can improve welfare by breaking self-reinforcing pessimism. In a Malthusian world with non-monotone dynamics, announcements require higher precision to avoid amplifying volatility.

Heterogeneity dampens over-coordination. In response to a public announcement, Romerians coordinate (amplifying the signal) while Malthusians anti-coordinate (dampening it). This heterogeneity reduces the risk of over-coordination on noise. The precision threshold for beneficial announcement is therefore lower in populations with substantial Malthusian minorities. Together with the equilibrium-structure argument of Section 4 (paragraph headed “The stabilizing role of Malthusians”), pessimists shrink the multiplicity region and dampen the response to noisy public signals, playing a stabilizing role on both margins.

7.6 Comparison of Decentralized and Planner Allocations

Table 7 summarizes the contrast across criteria.

Table 7. Decentralized vs. Planner Allocations by Welfare Criterion

	Dynastic	Average Util.	Total Util.
<i>Romerian World</i> ($\theta^* > 0$)			
Decentralized fertility		Too low (externality + wrong beliefs)	
Optimal announcement	Truthful	Mildly shaded toward θ_R	Strongly shaded toward θ_R
Selection-shifting role	None	Modest	Strongest
Residual inefficiency	Externality only	Reduced externality	Most reduced externality
<i>Malthusian World</i> ($\theta^* < 0$)			
Decentralized fertility		Too high (externality), mixed by beliefs	
Optimal announcement	Truthful	Mildly shaded toward θ_M	Strongly shaded toward θ_M
Selection-shifting role	None	Modest	Strongest
Residual inefficiency	Externality only	Reduced externality	Most reduced externality

Under all three criteria, the decentralized equilibrium is inefficient due to the uninternalized scale-effect externality. The dynastic planner can address belief distortions but not the externality. The total utilitarian planner shades further, reflecting an additional motive to value population. Achieving full efficiency would require additional instruments (fertility subsidies in Romerian worlds, fertility taxes in Malthusian worlds).

7.7 Policy Implications

The communication policy analyzed in this section can be understood as an *unconventional family policy*. Conventional family policy operates through fiscal instruments—childcare subsidies, baby bonuses, parental leave, child tax credits—that alter the price of children. The policy considered here operates instead by shaping beliefs about the long-run productivity consequences of population. The closest analogue is therefore not fiscal policy but central-bank communication, particularly the Delphic forward guidance studied by [Eggertsson and Woodford \(2003\)](#), [Woodford \(2005\)](#), [Blinder](#)

et al. (2008), and Campbell et al. (2012), which influences the economy by revealing information rather than by committing to future actions. As in monetary communication, credibility (τ_A) is essential, and optimal communication can depart from naive truth-telling once externalities are present (Theorem 5).

Within this analogy, the model yields a coherent set of practical lessons. The first concerns when to speak. When the economy appears stuck in the Malthusian trap, even a moderately imprecise signal may be worth announcing, because the welfare gain of escaping prolonged stagnation is large. Precisely when policymakers are least sure of what to say, the social cost of speaking imperfectly is lowest relative to the expected upside.

The second lesson concerns what to say, and here the welfare criterion is decisive. A planner who respects household preferences as given—the dynastic benchmark—should simply tell the truth: accurate information maximizes the value of the signal to the households who use it. A total utilitarian planner, by contrast, internalizes the externality that households ignore and should shade announcements systematically: toward optimism in Romerian worlds, where additional population produces under-counted social benefits, and toward pessimism in Malthusian worlds, where it imposes under-counted social costs. The magnitude of the optimal shading scales with the size of the externality, the deviation of current population from the reference level, and the credibility of the announcer.

Two further lessons emerge from the dynamics of the model. Near the multiplicity boundary, where ϕ is close to ϕ^{**} , the announcement does double duty: in addition to conveying information, it selects among multiple equilibria, and this selection-shifting role can be economically as important as the information-provision role. And during the non-monotone dynamics that arise in Malthusian regions, the timing of announcements matters in its own right. Speaking during an expansion—when population is high and signals are most informative—can accelerate learning and dampen the boom-bust swings that the model produces.¹⁷

The final and most important lesson is epistemic. Credibility is a stock variable: it must be built up during normal times so that announcements carry weight when intervention matters most. Credibility differs from knowledge, and the model is unsparing on this point. When the economy is in the trap, the planner knows the least about what to say, because the very signals that would identify the regime are weakest there. Acknowledging this uncertainty, rather than projecting false confidence, may be the most honest—and ultimately the most credible—policy.

¹⁷A further parallel deserves mention. Anticipated ex-post pronatalist support during fertility crises creates a time-inconsistency problem analogous to bailout policies. Anticipating support raises ex-ante fertility above the level optimal without commitment, amplifying cycle dynamics in a Malthusian world. Credible long-run population policy requires explicit commitment limits.

8 Conclusion

This paper develops an overlapping-generations framework in which households disagree about the long-run economic consequences of population, and traces the implications of that disagreement for fertility, growth, and learning. The mechanism is a feedback loop: worldviews about scale effects shape fertility decisions, fertility shapes the path of population, and population shapes the signals from which beliefs are subsequently learned. The framework yields a duality that organizes much of the analysis. In Romerian economies, Malthusian minorities can drag the economy toward stagnation; in Malthusian economies, Romerian minorities can generate the kind of non-monotone, boom–bust dynamics familiar from pre-industrial demography.

The object I find most distinctive is an *informational poverty trap*. It arises from the geometry of Bayesian learning rather than from any specific functional form: rival worldviews must agree at a common crossing point, so signals are least informative precisely in the region where pessimism keeps the economy confined. Beliefs that suppress fertility thus tend to insulate themselves from correction. This offers a way of rereading a long-standing puzzle in economic history. The pre-industrial era admits two interpretations within a single model—a genuine Malthusian regime that ended with a technological break, or a Romerian regime trapped for millennia by self-reinforcing pessimism that finally dissolved once population breached the informative region—and these two readings carry rather different implications for how one thinks about the modern fertility decline.

Taking belief heterogeneity seriously also changes how one thinks about policy. Government communication about long-run population consequences becomes an *unconventional family policy*, closer in spirit to central-bank forward guidance than to childcare subsidies or baby bonuses: a tool that operates through expectations rather than through fiscal instruments. Because the population over which welfare is evaluated is itself shaped by policy, dynastic, average-utilitarian, and total-utilitarian criteria can recommend systematically different communication strategies, which suggests that information and credibility deserve a place alongside conventional levers in the low-fertility policy toolkit. Many natural extensions remain—a fuller classification of Malthusian-world dynamics, sharper causal identification of the belief–fertility link, integration with the quantity–quality margin, a reputational microfoundation for announcement credibility, and a quantitative implementation that could begin to adjudicate between the two readings of pre-industrial history. I leave these possibilities to future research.

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A Proofs and Auxiliary Theoretical Results

A note on the appendix proofs

Throughout the appendix the law of motion of TFP is the general specification (3),

$$\ln A_{t+1} = \ln A_t + g(N_t; \theta^*) + \eta_{t+1}, \quad \eta_{t+1} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2),$$

with $g(\cdot; \theta)$ satisfying Assumption 1. I use the local elasticity notation

$$\tilde{\theta}_j \equiv g_N(\bar{N}; \theta_j) \bar{N},$$

which represents the local slope of $\ln(N/\bar{N}) \mapsto g(N; \theta_j)$ at the reference; under the leading log-linear specification $g(N; \theta) = \bar{g} + \theta \ln(N/\bar{N})$ this reduces to $\tilde{\theta}_j = \theta_j$, and the local linearization is exact. By (C4), $\text{sgn}(\tilde{\theta}_j) = \text{sgn}(\theta_j)$, so $\tilde{\theta}_R > 0 > \tilde{\theta}_M$ in any case.

The signal divergence used throughout is

$$\mathcal{I}_t = \frac{[g(N_t; \theta_R) - g(N_t; \theta_M)]^2}{2\sigma_\eta^2}, \quad (33)$$

which by (C1)–(C2) vanishes continuously as $N_t \rightarrow \bar{N}$ and by (C3) becomes arbitrarily large as $|\ln(N_t/\bar{N})|$ grows.

Several appendix results below rely on a local log-linear approximation of the household value function around a balanced-growth reference at $N = \bar{N}$. I distinguish between (a) parts of the argument that are exact under the general multiplicative log-normal technology of (3), and (b) parts that hold to first order around the reference. The Bayes-rule calculations for π_t and L_t are exact for any g satisfying (C1)–(C4); only the value-function expansion is approximated. The proof of Proposition 3 is a direct calculation that does not rely on a dynamical-systems reduction. The global-games selection of Theorem 2 adapts the supermodular-game machinery of Frankel et al. (2003) to the reduced Romerian subgame, with the existence step using the single-crossing argument of Athey (2001).

A.1 Proof of Lemma 1

Step 1: Bellman equation. Substituting $c_{j,t} = A_t(1 - \psi(n))$ and using $\ln c_{j,t} = \ln A_t + \ln(1 - \psi(n))$ into the dynastic objective (7) yields the Bellman equation (16):

$$V_j(A, N, \phi, \pi) = \max_{n \geq 0} \left\{ \ln A + \ln(1 - \psi(n)) + \beta n \mathbb{E}_j[V_j(A', N', \phi', \pi')] \right\}.$$

The choice n enters the period return only through $\ln(1 - \psi(n))$ and the discount-multiplier βn ; aggregate states enter parametrically.

Step 2: Local log-linear conjecture. I conjecture

$$V_j(A, N, \phi, \pi) = v_j(\phi, \pi) + \xi_{A,j} \ln A + \xi_{N,j} \ln(N/\bar{N}) + R_j(A, N, \phi, \pi), \quad (34)$$

where R_j is a remainder of order $O(\|(\ln A - \ln A^*, \ln(N/\bar{N}))\|^2)$ around the reference $(A^*, \bar{N})$. The technology side of the Bellman is exactly linear under (3) after replacing $g(N; \theta)$ by its first-order Taylor expansion around $\bar{N}$:

$$g(N; \theta) = \bar{g} + \tilde{\theta} \ln(N/\bar{N}) + r(N; \theta), \quad r(N; \theta) = O(\ln(N/\bar{N})^2), \quad (35)$$

with $\tilde{\theta} = g_N(\bar{N}; \theta)\bar{N}$ and r absorbed into R_j . Under the log-linear leading case $r \equiv 0$ and the conjecture is exact in $\ln(N/\bar{N})$.

Step 3: Expected continuation under belief θ_j . From (3) and (35),

$$\mathbb{E}_j[\ln A'] = \ln A + \bar{g} + \tilde{\theta}_j \ln(N/\bar{N}) + O(\ln(N/\bar{N})^2),$$

since $\mathbb{E}_j[\eta_{t+1}] = 0$. For population, with $N' = \bar{n}^e N$,

$$\mathbb{E}_j[\ln(N'/\bar{N})] = \ln(N/\bar{N}) + \ln \bar{n}^e.$$

Substituting into (34),

$$\mathbb{E}_j[V_j'] = \mathbb{E}_j v_j(\phi', \pi') + \xi_{A,j} [\ln A + \bar{g} + \tilde{\theta}_j \ln(N/\bar{N})] + \xi_{N,j} [\ln(N/\bar{N}) + \ln \bar{n}^e] + O(\sigma_\eta^2) + O(\ln(N/\bar{N})^2), \quad (36)$$

where the $O(\sigma_\eta^2)$ term is a Jensen correction absorbed into v_j .

Step 4: Coefficient matching. Substituting (36) into (16) at the optimum n_j^* (treated as locally constant in $(\ln A, \ln(N/\bar{N}))$) and matching coefficients on $\ln A$:

$$\xi_{A,j} = 1 + \beta n_j^* \xi_{A,j} \implies \xi_{A,j} = \frac{1}{1 - \beta n_j^*}.$$

Matching coefficients on $\ln(N/\bar{N})$:

$$\xi_{N,j} = \beta n_j^* (\xi_{A,j} \tilde{\theta}_j + \xi_{N,j}) \implies \xi_{N,j} (1 - \beta n_j^*) = \frac{\beta n_j^* \tilde{\theta}_j}{1 - \beta n_j^*},$$

yielding

$$\xi_{N,j} = \frac{\beta n_j^* \tilde{\theta}_j}{(1 - \beta n_j^*)^2}.$$

This is (19). The constant term v_j absorbs $\bar{g}\xi_{A,j}$, $\xi_{N,j} \ln \bar{n}^e$, the Jensen term, and $\beta n_j^* \mathbb{E}_j v_j'$. The condition $\beta n_j^* < 1$ ensures finiteness.

Step 5: Fertility FOC. Under $\psi(n) = 1 - e^{-(\gamma/2)n^2}$, $\psi'(n)/(1 - \psi(n)) = \gamma n$, and the FOC of (16) reads

$$\gamma n_j^* = \beta \mathbb{E}_j[V_j']. \quad (37)$$

Step 6: Existence and uniqueness of the fixed point. Substituting (36) into (37), the right-hand side defines a continuous function $F(n_j^*)$ on $[0, 1/\beta)$ with $F(0)$ finite and $F(n) \rightarrow \infty$ as $n \uparrow 1/\beta$. The map $n \mapsto \gamma n - F(n)$ admits a unique root in $[0, \bar{n}^{\max}]$ for some $\bar{n}^{\max} < 1/\beta$.

Step 7: Comparative statics in θ_j . Differentiating (37) totally with respect to θ_j at fixed reference $(\bar{n}, A^*, N^*, \phi, \pi)$ and $\ln(N^*/\bar{N}) = 0$, the only explicit- θ_j channel is through $\xi_{N,j}$, since $\xi_{A,j}$ does not depend on θ_j and the term $\xi_{A,j} \tilde{\theta}_j \ln(N^*/\bar{N})$ vanishes at the reference:

$$\gamma \left. \frac{\partial n_j^*}{\partial \theta_j} \right|_{\text{explicit}} = \beta \frac{\partial \xi_{N,j}}{\partial \tilde{\theta}_j} \cdot \frac{d\tilde{\theta}_j}{d\theta_j} = \frac{\beta^2 n_j^*}{(1 - \beta n_j^*)^2} \cdot \frac{d\tilde{\theta}_j}{d\theta_j}.$$

By (C4), $d\tilde{\theta}_j/d\theta_j > 0$ at $\bar{N}$, so the explicit channel is strictly positive. Implicit terms (in n_j^*) generate a multiplier $(1 - \beta \partial F/\partial n)^{-1}$ which is positive under $\beta n_j^* < 1$, so the total derivative $\partial n_j^*/\partial \theta_j > 0$. Writing $n_j^* = n^* h(\theta_j; \bar{n})$ with h increasing in θ_j follows by reorganization. In particular, since $\theta_R > 0 > \theta_M$ and (C4) implies $\tilde{\theta}_R > 0 > \tilde{\theta}_M$, we have $n_R^* > n_M^*$. $\square$

A.2 Proof of Proposition 1

For type j expecting aggregate fertility $\bar{n}^e$, (36) gives

$$\mathbb{E}_j[V_j'] = v_j^{\text{const}} + \xi_{A,j} \ln A + \xi_{A,j} \tilde{\theta}_j \ln(N/\bar{N}) + \xi_{N,j} \ln(N/\bar{N}) + \xi_{N,j} \ln \bar{n}^e + O(\ln(N/\bar{N})^2),$$

so

$$\frac{\partial \mathbb{E}_j[V'_j]}{\partial \bar{n}^e} = \frac{\xi_{N,j}}{\bar{n}^e}.$$

Differentiating the FOC $\gamma n_j^* = \beta \mathbb{E}_j[V'_j]$ totally with respect to $\bar{n}^e$ and using the closed-form for $\xi_{N,j}$:

$$\left. \frac{dn_j^*}{d\bar{n}^e} \right|_{\text{ref}} = \frac{\beta \xi_{N,j}}{\gamma \bar{n}^e} = \frac{\beta^2 n_j^* \tilde{\theta}_j}{\gamma \bar{n}^e (1 - \beta n_j^*)^2}.$$

This is (20). The sign matches the sign of $\tilde{\theta}_j$, which by (C4) matches the sign of θ_j : $d_R > 0$ (complementarity) and $d_M < 0$ (substitutability). $\square$

A.3 Proof of Theorem 1

Equilibrium condition. Let

$$BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1 - \phi) n_M^*(\bar{n}^e).$$

A rational-expectations equilibrium is a fixed point $\bar{n}^* = BR(\bar{n}^*)$. The local slope is

$$\rho(\phi) = BR'(\bar{n}^*) = \phi d_R + (1 - \phi) d_M,$$

with $d_R > 0$ and $d_M < 0$ by Proposition 1. The argument depends on g only through the signs of $\tilde{\theta}_R$ and $\tilde{\theta}_M$, guaranteed by (C4).

Part (i): Malthusian-dominated region ($\phi < \phi^*$). For $\phi < \phi^* = |d_M|/(d_R + |d_M|)$, $\rho(\phi) < 0$. Define $\Phi(x) = BR(x) - x$, continuously differentiable with $\Phi'(x) = \rho(\phi) - 1 < -1 < 0$ at any fixed point. By the implicit function theorem, fixed points are isolated. Since $n_R^*(\bar{n}^e)$ and $n_M^*(\bar{n}^e)$ are continuous and bounded above by some $\bar{n}^{\max} < 1/\beta$, BR maps $[0, \bar{n}^{\max}]$ into itself; by the intermediate value theorem applied to Φ , at least one fixed point exists. Strict monotonicity of Φ implies it is unique. The argument requires only the sign of $\rho(\phi)$.

Part (ii): Contraction region. For ϕ in this range, $0 \leq \rho(\phi) < 1$. The map $BR : [0, \bar{n}^{\max}] \rightarrow [0, \bar{n}^{\max}]$ is Lipschitz with constant $\rho(\phi) < 1$ on a neighborhood of any candidate fixed point. By Banach's fixed-point theorem, the fixed point is unique. When $d_R \leq 1$, $\phi^{**} \geq 1$, so the contraction region extends to all $\phi \in [\phi^*, 1]$.

Part (iii): Multiplicity region ($\phi > \phi^{}$; requires $d_R > 1$).** For $\phi > \phi^{**}$, $\rho(\phi) > 1$. Locally near any fixed point, $\Phi'(x) = \rho - 1 > 0$, so BR crosses the 45-degree line from below. Since BR is continuous on $[0, \bar{n}^{\max}]$ with $BR(0) \geq 0$ and $BR(\bar{n}^{\max}) \leq \bar{n}^{\max}$, Φ takes nonnegative values

near 0 and nonpositive near $\bar{n}^{\max}$. By the intermediate value theorem combined with the local slope condition, the number of transversal intersections is odd and at least three. Standard transversality arguments imply local slopes alternate, so the extreme intersections are stable.

Multiplicity condition. $\phi^{**} < 1$ iff $d_R > 1$. Substituting (20):

$$\frac{\beta^2 n_R^* \tilde{\theta}_R}{\gamma \bar{n} (1 - \beta n_R^*)^2} > 1 \iff \tilde{\theta}_R > \tilde{\theta}_R^{\text{crit}} \equiv \frac{\gamma \bar{n} (1 - \beta n_R^*)^2}{\beta^2 n_R^*}. \quad \square$$

A.4 Proof of Theorem 2

I adapt the supermodular global-games argument of Frankel et al. (2003) via a reduction that substitutes out the Malthusian best response. The existence step uses the single-crossing approach of Athey (2001).

Step 1: Reduction to a Romerian subgame. Let n_R^i denote individual Romerian fertility choices and $\bar{n}_R = \int_R n_R^i di / \phi$ average Romerian fertility. The aggregate fertility identity is

$$\bar{n} = \phi \bar{n}_R + (1 - \phi) n_M^*(\bar{n}).$$

Because $d_M < 0$ by Proposition 1 (which uses only (C4)), the map $\bar{n} \mapsto \phi \bar{n}_R + (1 - \phi) n_M^*(\bar{n})$ for fixed $\bar{n}_R$ has derivative $(1 - \phi) d_M < 0$, and so the equation has a unique solution $\bar{n} = T(\bar{n}_R)$. By the implicit function theorem,

$$T'(\bar{n}_R) = \frac{\phi}{1 - (1 - \phi) d_M} > 0.$$

The reduced game has Romerian players choosing n_R^i with payoffs $\tilde{U}_R(n_R^i, \bar{n}_R) := U_R(n_R^i, T(\bar{n}_R))$.

Step 2: Supermodularity of the reduced game.

$$\frac{\partial^2 \tilde{U}_R}{\partial n_R^i \partial \bar{n}_R} = \frac{\partial^2 U_R}{\partial n_R^i \partial \bar{n}} \cdot T'(\bar{n}_R).$$

By Proposition 1, the cross-partial $\partial^2 U_R / \partial n_R^i \partial \bar{n}$ has the sign of $\tilde{\theta}_R$, which is positive by (C4). Combined with $T' > 0$, the reduced game is strictly supermodular among Romerians.

Step 3: Monotone signal structure. Each Romerian observes $\phi_i = \phi + \zeta_i$ with $\zeta_i \sim \mathcal{N}(0, \sigma_\zeta^2)$ i.i.d. With a diffuse improper prior on ϕ , the conditional distribution of ϕ given ϕ_i is Gaussian with mean ϕ_i and variance σ_ζ^2 , satisfying the monotone likelihood ratio property. The reduced game now

satisfies the conditions of Theorem 1 of [Athey \(2001\)](#): payoff differences between any two actions are single-crossing in the type, the type space is ordered, and posteriors satisfy MLRP. Athey's theorem guarantees existence of monotone pure-strategy equilibria.

Step 4: Dominance regions. Under the calibration of the multiplicity region, there exist values $\bar{\phi}_L < \phi^{**}$ and $\bar{\phi}_H > 1$ such that for $\phi < \bar{\phi}_L$ the low-fertility selection strictly dominates regardless of others' actions (Malthusians outweigh Romerians), and for $\phi > \bar{\phi}_H$ the high-fertility selection strictly dominates (continuation value of children under (C4) exceeds the time cost at any aggregate). Both follow by direct evaluation of the FOC at the extreme aggregate fertility levels.

Step 5: Application of [Frankel et al. \(2003\)](#). The reduced Romerian game satisfies the maintained conditions of Theorem 1 of [Frankel et al. \(2003\)](#): (a) supermodular payoffs in actions; (b) state-monotone payoffs (since $\partial U_R / \partial \bar{n}_R > 0$ under (C4)); (c) continuous payoffs; (d) dominance regions on both ends. The theorem therefore applies: as $\sigma_\zeta \rightarrow 0$, iterated deletion of strictly dominated strategies converges to a unique strategy profile in monotone (threshold) form.

Step 6: Closed-form risk-dominant cutoff. At $\phi_i = \phi^{rd}$, a Romerian must be indifferent between the two selections under Laplacian beliefs. The indifference condition reduces to

$$\int_0^1 [U_R(n_R^*(\bar{n}_q), \bar{n}_q) - U_R^0] dq = 0,$$

where $\bar{n}_q$ is the aggregate fertility implied by a fraction q of Romerians playing high. Equivalently,

$$\int_{\bar{n}^L}^{\bar{n}^H} u'_R(n_R^*(\bar{n}^e)) d\bar{n}^e = 2 \int_{\bar{n}^L}^{\bar{n}^{rd}} u'_R(n_R^*(\bar{n}^e)) d\bar{n}^e,$$

which under the log-quadratic specification has a unique closed-form solution $\bar{n}^{rd}$, and ϕ^{rd} is recovered from $T^{-1}(\bar{n}^{rd}) = \phi \bar{n}_R^{rd} + (1 - \phi)n_M^*(\bar{n}^{rd})$. The selected equilibrium is risk-dominant in the sense of [Frankel et al. \(2003\)](#). $\square$

A.5 Proof of Theorem 3

Step 1: Demographic-selection drift. For interior ϕ_t ,

$$\phi_t^{birth} - \phi_t = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t} > 0,$$

where $n_R > n_M$ by Lemma 1 (using (C4)). The selection drift is strictly positive at every interior ϕ_t with no information dependence.

Step 2: Posterior dynamics under truth. Let $L_t = \ln[(1 - \pi_t)/\pi_t]$. Bayes' rule (12) together with the Gaussian likelihoods (11) yields, for any g satisfying Assumption 1,

$$L_{t+1} - L_t = -\frac{g(N_t; \theta_R) - g(N_t; \theta_M)}{\sigma_\eta^2} \left[s_t - \frac{1}{2}(g(N_t; \theta_R) + g(N_t; \theta_M)) \right], \quad (38)$$

which follows from expanding the log-likelihood ratio of two Gaussian densities with means $g(N_t; \theta_R)$ and $g(N_t; \theta_M)$ and common variance σ_η^2 . Under truth $\theta^* = \theta_R$, $\mathbb{E}^{\theta_R}[s_t | \mathcal{F}_t] = g(N_t; \theta_R)$, so

$$\mathbb{E}^{\theta_R}[L_{t+1} - L_t | \mathcal{F}_t] = -\frac{[g(N_t; \theta_R) - g(N_t; \theta_M)]^2}{2\sigma_\eta^2} = -\mathcal{I}_t \leq 0.$$

Therefore L_t is a supermartingale under truth. By Doob's martingale convergence theorem, L_t converges a.s. to some $[-\infty, \infty]$ -valued limit.

Step 3: $\sum_t \mathcal{I}_t = \infty$ almost surely. Suppose, for contradiction, that $\sum_t \mathcal{I}_t < \infty$ on a set of positive probability. Then $\mathcal{I}_t \rightarrow 0$, which by (C2) and (C3) implies $N_t \rightarrow \bar{N}$ (the only zero of $\mathcal{I}$), so $\bar{n}_t \rightarrow 1$ along a subsequence of full asymptotic density.

By Assumption 5, $\bar{n}_t \rightarrow 1$ requires $\phi_t \rightarrow \phi^{trap}$. On the same event, the cumulative posterior increments are summable (since $|\Delta\pi_t| = O(|\Delta L_t|)$ and $\mathbb{E}|\Delta L_t| = O(\sqrt{\mathcal{I}_t})$ by Cauchy-Schwarz, with $\sum \sqrt{\mathcal{I}_t} < \infty$ under $\sum \mathcal{I}_t < \infty$ and the additional regularity in the trap). Through (15), the cumulative cultural-transmission drift accumulates:

$$\phi_{t+1} - \phi_t \geq \lambda(\phi_t^{birth} - \phi_t) - (1 - \lambda)|\Delta\pi_{t+1}|,$$

with $\sum |\Delta\pi_{t+1}| < \infty$. Selection drift is bounded below by λc for some $c > 0$ along the subsequence (since ϕ_t bounded away from 0 and 1). Summing, ϕ_t increases beyond any bound below 1, contradicting $\phi_t \rightarrow \phi^{trap} < 1$. Hence $\sum_t \mathcal{I}_t = \infty$ a.s.

Step 4: $\pi_t \rightarrow 1$ almost surely. Combining $\mathbb{E}^{\theta_R}[\Delta L_t] = -\mathcal{I}_t$ with $\sum \mathcal{I}_t = \infty$ and the law of large numbers for martingale differences (the variance of ΔL_t is also $\mathcal{I}_t$ -bounded), $L_t \rightarrow -\infty$ a.s., hence $\pi_t \rightarrow 1$ a.s.

Step 5: $\phi_t \rightarrow 1$ almost surely. Given $\pi_t \rightarrow 1$, for any $\varepsilon > 0$ there exists $T(\omega)$ with $\pi_t > 1 - \varepsilon$ for $t \geq T$. The dynamics (15) give

$$1 - \phi_{t+1} \leq \lambda(1 - \phi_t) + (1 - \lambda)\varepsilon,$$

where the inequality uses $\phi_t^{birth} \geq \phi_t$ (Step 1). Iterating, $\limsup(1 - \phi_t) \leq \varepsilon$. Since ε is arbitrary, $\phi_t \rightarrow 1$ a.s.

Step 6: $\bar{n}_t \rightarrow n_R^*$ and $\sup_t \ln(N_t/\bar{N}) = +\infty$. Convergence of fertility follows by continuity. Since $n_R^* > 1$, eventually $\bar{n}_t \geq (1 + n_R^*)/2 > 1$, so N_t grows geometrically and $\sup_t \ln(N_t/\bar{N}) = +\infty$. $\square$

A.6 Proof of Proposition 2

Throughout, the trajectory is assumed to remain in $\mathcal{T}(\delta, \zeta)$; the proof also verifies trap closure. The proof uses only (C1)–(C2).

Part (i): Bound on $\mathbb{E}[\Delta\pi_t \mid \mathcal{F}_t]$. From $\pi_t = (1 + e^{L_t})^{-1}$, $d\pi/dL = -\pi(1 - \pi)$. A second-order Taylor expansion gives

$$\Delta\pi_t = -\pi_t(1 - \pi_t)\Delta L_t + \frac{1}{2}\pi_t(1 - \pi_t)(1 - 2\pi_t)(\Delta L_t)^2 + O((\Delta L_t)^3).$$

Taking expectations under truth and using $\mathbb{E}[(\Delta L_t)^2 \mid \mathcal{F}_t] = O(\mathcal{I}_t)$ (from (38), the conditional variance of ΔL_t is $\mathcal{I}_t/(2\sigma_\eta^2) \cdot 2\sigma_\eta^2 = 2\mathcal{I}_t$):

$$|\mathbb{E}[\Delta\pi_t \mid \mathcal{F}_t]| \leq \pi_t(1 - \pi_t)\mathcal{I}_t + O(\mathcal{I}_t).$$

In the trap region, $|\ln(N_t/\bar{N})| < \delta$, and by (C1)–(C2), $\mathcal{I}_t \leq \bar{\mathcal{I}}(\delta)$ where $\bar{\mathcal{I}}(\delta) = \sup_{|N/\bar{N}-1|<\delta'} [g(N; \theta_R) - g(N; \theta_M)]^2/(2\sigma_\eta^2) \rightarrow 0$ as $\delta \rightarrow 0$. Since $\pi(1 - \pi) \leq 1/4$, the leading term is bounded by $\bar{\mathcal{I}}(\delta)/4 = C_1\bar{\mathcal{I}}(\delta)$ with $C_1 = 1/4$.

Part (ii): Bound on demographic-selection drift.

$$\phi_{t+1}^{birth} - \phi_t = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t} \leq \frac{(1 - \phi_t)(n_R - n_M)}{n_M},$$

where the inequality uses $\phi_t/\bar{n}_t \leq 1/n_M$ (since $\bar{n}_t \geq n_M$). Therefore $C_2 = 1/n_M$.

Trap closure. If $(\phi_t, N_t, \pi_t) \in \mathcal{T}(\delta, \zeta)$, then $(\phi_{t+1}, N_{t+1}, \pi_{t+1}) \in \mathcal{T}(\delta, \zeta)$ for δ, ζ small enough. The condition $\bar{n}(\phi_{t+1}) \in [1, 1 + \zeta]$ follows from continuity of $\bar{n}$ in ϕ and the small change in ϕ per period (Parts (i)–(ii)). The condition $|\ln(N_{t+1}/\bar{N})| < \delta$ follows from $\bar{n}_t \in [1, 1 + \zeta]$ provided ζ is small enough relative to δ . The condition $\pi_{t+1} \leq \phi_{t+1}$ follows from

$$\phi_{t+1} - \pi_{t+1} = \lambda\phi_t^{birth} + (1 - \lambda)\pi_{t+1} - \pi_{t+1} = \lambda(\phi_t^{birth} - \pi_{t+1}),$$

and $\phi_t^{birth} \geq \phi_t \geq \pi_t$ by hypothesis, while in expectation under truth $\pi_{t+1} \geq \pi_t$ (Step 2 of Theorem 3); for the realized trajectory, the trap is closed in expectation, with deviations controlled by the bounded variance of $\Delta\pi_t$.

Part (iii): Lower bound on $\mathbb{E}[T^{esc}]$. Combining (i) and (ii),

$$\mathbb{E}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t] \leq \lambda C_2(n_R - n_M)(1 - \phi_t) + (1 - \lambda)C_1\bar{\mathcal{I}}(\delta) \leq D.$$

Define $M_t = \phi_t - tD$. Then M_t is a supermartingale on the trap event. Optional stopping applied to $T \wedge n$:

$$D \mathbb{E}[T \wedge n] \geq \bar{\phi} \mathbb{P}(T \leq n) - \phi_0.$$

Letting $n \rightarrow \infty$,

$$\mathbb{E}[T^{esc}] \geq \frac{\bar{\phi} - \phi_0}{D}. \quad \square$$

A.7 Trap Duration (Auxiliary Result)

Corollary 3. *Under Proposition 2, suppose $g(N; \theta_R) - g(N; \theta_M) = O(|\ln(N/\bar{N})|^k)$ near $\bar{N}$ for some $k \geq 1$. When learning is the binding escape channel, $\mathbb{E}[T^{esc}] = \Omega(\delta^{-2k})$ as $\delta \rightarrow 0$. Under the log-linear leading case $k = 1$ so $\mathbb{E}[T^{esc}] = \Omega(\delta^{-2})$.*

Proof. By hypothesis, $\bar{\mathcal{I}}(\delta) = O(\delta^{2k})$. With $D = a + b\delta^{2k}$, we have $\mathbb{E}[T^{esc}] \geq (\bar{\phi} - \phi_0)/(a + b\delta^{2k})$. When the cultural channel is shut down or weak relative to the learning channel, a is small and $\mathbb{E}[T^{esc}] \geq \Omega(\delta^{-2k})$. $\square$

A.8 Trap Boundary (Auxiliary Result)

Proposition 7. *The critical population at which the learning channel matches the demographic-selection channel in magnitude solves*

$$[g(N^{crit}; \theta_R) - g(N^{crit}; \theta_M)]^2 = \frac{2\sigma_\eta^2 \cdot \lambda C_2(n_R - n_M)(1 - \phi)}{(1 - \lambda)C_1}.$$

Under the leading log-linear specification $g(N; \theta) = \bar{g} + \theta \ln(N/\bar{N})$, this becomes

$$\frac{N^{crit}(\phi)}{\bar{N}} = \exp\left(\sqrt{\frac{2\sigma_\eta^2 \cdot \lambda C_2(n_R - n_M)(1 - \phi)}{(1 - \lambda)C_1(\theta_R - \theta_M)^2}}\right).$$

Proof. Equate the per-period selection drift bound (Proposition 2(ii)) with the per-period learning drift bound (Proposition 2(i)) using $\mathcal{I} = [g(N; \theta_R) - g(N; \theta_M)]^2/(2\sigma_\eta^2)$, and solve for N . Under

the log-linear case, $g(N; \theta_R) - g(N; \theta_M) = (\theta_R - \theta_M) \ln(N/\bar{N})$, giving the explicit form. $\square$

A.9 Proof of Proposition 3

Part (i): Selection drift on births.

$$\phi_t^{birth} - \phi_t = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t}.$$

Since $\phi_t \in (0, 1)$ implies $\phi_t(1 - \phi_t) \in (0, 1/4]$, and $\bar{n}_t \geq n_M > 0$, the drift is strictly positive and bounded above by $(n_R - n_M)/(4n_M)$, independently of N_t and of the form of g beyond what guarantees $n_R > n_M$ (Lemma 1 under (C4)).

Part (ii): Learning drift on the posterior. Repeating (38) under truth $\theta^* = \theta_M$, $\mathbb{E}^{\theta_M}[s_t | \mathcal{F}_t] = g(N_t; \theta_M)$. Substituting,

$$\mathbb{E}^{\theta_M}[L_{t+1} - L_t | \mathcal{F}_t] = -\frac{g(N_t; \theta_R) - g(N_t; \theta_M)}{\sigma_\eta^2} \cdot \frac{-(g(N_t; \theta_R) - g(N_t; \theta_M))}{2} = +\mathcal{I}_t.$$

Using the chain-rule expansion $d\pi/dL = -\pi(1 - \pi)$,

$$\mathbb{E}^{\theta_M}[\pi_{t+1} - \pi_t | \mathcal{F}_t] = -\pi_t(1 - \pi_t)\mathcal{I}_t + o(\mathcal{I}_t).$$

By (C1), $\mathcal{I}_t = 0$ at $N_t = \bar{N}$; by (C3), $\mathcal{I}_t$ becomes arbitrarily large as N_t deviates substantially from $\bar{N}$. The drift is therefore strictly negative for $\pi_t \in (0, 1)$ and $N_t \neq \bar{N}$, and exactly zero at $N_t = \bar{N}$.

Part (iii): Sign change of the joint drift in ϕ_t . From $\phi_{t+1} = \lambda\phi_t^{birth} + (1 - \lambda)\pi_{t+1}$,

$$\phi_{t+1} - \phi_t = \lambda(\phi_t^{birth} - \phi_t) + (1 - \lambda)(\pi_t - \phi_t) + (1 - \lambda)(\pi_{t+1} - \pi_t).$$

Taking expectations under truth and using Part (ii):

$$\mathbb{E}^{\theta_M}[\phi_{t+1} - \phi_t | \mathcal{F}_t] = \underbrace{\lambda(\phi_t^{birth} - \phi_t)}_{>0} + \underbrace{(1 - \lambda)(\pi_t - \phi_t)}_{\text{sign of } \pi_t - \phi_t} + \underbrace{(1 - \lambda)[- \pi_t(1 - \pi_t)\mathcal{I}_t + o(\mathcal{I}_t)]}_{\leq 0}. \quad (39)$$

At $N_t = \bar{N}$: $\mathcal{I}_t = 0$ by (C1), so the third term vanishes. The drift reduces to the first two terms and is non-negative whenever $\pi_t \geq \phi_t - \frac{\lambda}{1-\lambda}(\phi_t^{birth} - \phi_t)$.

For N_t far from $\bar{N}$: by (C3), $\mathcal{I}_t$ becomes arbitrarily large, while the first two terms remain

bounded. The third term dominates and the drift is strictly negative once

$$\pi_t(1 - \pi_t)\mathcal{I}_t > \frac{\lambda(\phi_t^{birth} - \phi_t)}{1 - \lambda} + \max\{\pi_t - \phi_t, 0\}.$$

Substituting $\mathcal{I}_t = [g(N_t; \theta_R) - g(N_t; \theta_M)]^2 / (2\sigma_\eta^2)$ and solving:

$$[g(N_t; \theta_R) - g(N_t; \theta_M)]^2 > \frac{2\sigma_\eta^2 [\lambda(\phi_t^{birth} - \phi_t)/(1 - \lambda) + \max\{\pi_t - \phi_t, 0\}]}{\pi_t(1 - \pi_t)}.$$

By (C2)–(C3), the left-hand side is continuous in N_t , equals zero at $N_t = \bar{N}$, and grows without bound as N_t deviates substantially from $\bar{N}$. By the intermediate value theorem, there exists a population level at which the drift switches sign. Under the leading log-linear case, the threshold simplifies to

$$[\ln(N_t/\bar{N})]^2 > \bar{L}(\phi_t, \pi_t, \lambda)^2 \equiv \frac{2\sigma_\eta^2 [\lambda(\phi_t^{birth} - \phi_t)/(1 - \lambda) + \max\{\pi_t - \phi_t, 0\}]}{\pi_t(1 - \pi_t)(\theta_R - \theta_M)^2}. \quad \square$$

A.10 State-Dependent Value of Announcement (Auxiliary Results)

Proposition 8. *Suppose the economy is in $\mathcal{T}(\delta, \zeta)$ in a Romerian world. Under W^{tot} ,*

$$\mathcal{V}^{trap}(\tau_\nu^{trap}) = P^{esc}(\tau_\nu^{trap}) \cdot \Delta W^{esc} - P^{deepen}(\tau_\nu^{trap}) \cdot \Delta W^{deepen},$$

with a threshold $\underline{\tau}^{trap}$ above which $\mathcal{V}^{trap} > 0$. Because ΔW^{esc} is large, $\underline{\tau}^{trap}$ is moderate.

Sketch of proof. An announcement of precision τ_ν shifts the social posterior π_t^{post} toward the announced value, with magnitude proportional to $\tau_\nu/(\tau_\nu + \tau_{prior})$, and through (15) shifts the next-generation type share ϕ_{t+1} . In the trap, ϕ_t is below ϕ^{trap} . A truthful announcement raises π_t^{post} toward Romerianism, which through recruitment raises ϕ_{t+1} and hence $\bar{n}_{t+1}$. If the resulting $\bar{n}_{t+1}$ exceeds 1, population grows, learning engages by (C3), and escape probability P^{esc} becomes positive. The deepening event (announcement noise pushing π_t^{post} further toward Malthusianism) has probability bounded above by $\mathbb{P}(\nu < -\nu_{min})$, small for moderate τ_ν . The threshold $\underline{\tau}^{trap}$ is moderate because ΔW^{esc} reflects the welfare gain of escaping prolonged stagnation. $\square$

Proposition 9. *In a Malthusian world with non-monotone dynamics under W^{tot} ,*

$$\mathcal{V}^{cycle}(\tau_\nu) = \Omega(\text{Var}(\theta_i) - \mathbb{E}[\text{Var}(\theta_i^{post})]) - \Lambda((\bar{\theta} - \theta^*)^2 - \mathbb{E}[(\bar{\theta}^{post} - \theta^*)^2]) - C^{adj},$$

where $\Omega > 0$ is the marginal welfare gain from reducing dispersion, $\Lambda > 0$ from reducing mean bias, and $C^{adj} > 0$ is an adjustment cost. For sufficiently precise signals, $\mathcal{V}^{cycle} > 0$.

Sketch of proof. The decomposition (30) provides the mapping from variance and bias of beliefs to welfare losses Δ_3 and Δ_2 . Both variance and bias are decreasing in τ_ν . The adjustment cost reflects the welfare loss from disrupting agents' contingent plans during a cycle. $\square$

B Welfare Analysis Proofs and Auxiliary Results

B.1 Proof of Proposition 4

Step 1: Planner's Bellman. Define

$$V^P(A_t, N_t) = \max_{n_t} \left\{ N_t [\ln A_t + \ln(1 - \psi(n_t))] + \beta \mathbb{E}^*[V^P(A_{t+1}, n_t N_t)] \right\}.$$

Conjecture $V^P(A, N) = N \cdot w^P(A, N)$ with $w^P(A, N) \approx v^P + \xi_A^P \ln A + \xi_N^P \ln(N/\bar{N})$. Substituting and dividing by N ,

$$w^P(A, N) = \ln A + \ln(1 - \psi(n)) + \beta n \mathbb{E}^*[w^P(A', N')].$$

The planner's per-capita Bellman has the same form as the household's, but the planner additionally internalizes the population-multiplier through $N_{t+1} = nN_t$.

Step 2: Coefficient matching. Using $\mathbb{E}^*[\ln A'] = \ln A + \bar{g} + \tilde{\theta}^* \ln(N/\bar{N}) + O(\ln(N/\bar{N})^2)$ under (3) with the local Taylor expansion (35):

$$\xi_A^P = \frac{1}{1 - \beta n^{FB}}, \quad \xi_N^P = \frac{\beta n^{FB} \tilde{\theta}^*}{(1 - \beta n^{FB})^2}.$$

The sign of ξ_N^P matches the sign of $\tilde{\theta}^*$, which by (C4) matches the sign of θ^* .

Step 3: Planner FOC. Differentiating with respect to n_t and using $N_{t+1} = n_t N_t$, at the reference $\partial w^P / \partial N = \xi_N^P / N$, so the marginal value of population is $V_N^P = w^P + \xi_N^P$. The FOC reads (29).

Step 4: Comparison with household. The household's common-correct-belief FOC is $\gamma n^{CB} = \beta \mathbb{E}[V_{CB}(A', N')]$ where $V_{CB}(A, N) = w^P(A, N)$. The wedge between (29) and the household FOC is exactly $\beta \xi_N^P$, with the sign of θ^* by (C4). Strict monotonicity of γn gives

$$\theta^* > 0 \implies n^{FB} > n^{CB}, \quad \theta^* < 0 \implies n^{FB} < n^{CB}.$$

Step 5: $n^{CB} > n_M^*$. By Lemma 1, fertility is increasing in θ_j . With $\theta^* = \theta_R > 0$, $n^{CB} = n^*(\theta_R) > n_M^* = n^*(\theta_M)$. Combined with Step 4, $n^{FB} > n^{CB} > n_M^*$.

Step 6: Welfare decomposition. Established below in the proof of Proposition 5. $\square$

B.2 First-Best under Average Utilitarianism (Auxiliary Result)

Proposition 10. *In a Romerian world, $n^{CB} < n^{FB,avg} < n^{FB,tot}$. The average-utilitarian planner internalizes the per-capita scale-effect channel through $\ln A_t$, but lacks the population-multiplier term N_t that gives the total utilitarian planner the additional wedge.*

Proof. The average-utilitarian Bellman is $V^{P,avg}(A, N) = \ln A + \ln(1 - \psi(n)) + \beta \mathbb{E}^*[V^{P,avg}(A', N')]$, without the multiplicative N . Matching coefficients under the local linearization gives $\xi_N^{P,avg} = \beta n^{FB,avg} \tilde{\theta}^* / (1 - \beta n^{FB,avg})^2$. The wedge under total utilitarianism is $\beta(w^P + \xi_N^P)$ and under average utilitarianism is $\beta \xi_N^{P,avg}$. With $w^P > 0$ at the reference, $\beta \xi_N^P + \beta w^P > \beta \xi_N^{P,avg} > 0$ under $\tilde{\theta}^* > 0$. Strict monotonicity of γn gives $n^{FB,tot} > n^{FB,avg} > n^{CB}$. $\square$

B.3 Proof of Proposition 5

Decompose by inserting two intermediate allocations:

$$W^{FB}(\theta^*) - W^{HB} = \underbrace{[W^{FB}(\theta^*) - W^{CB}(\theta^*)]}_{\Delta_1} + \underbrace{[W^{CB}(\theta^*) - W^{CB}(\bar{\theta})]}_{\Delta_2} + \underbrace{[W^{CB}(\bar{\theta}) - W^{HB}]}_{\Delta_3},$$

where $W^{CB}(\theta)$ is welfare under decentralized common-belief equilibrium when households all believe θ ; W^{HB} is heterogeneous-beliefs welfare.

Δ_1 (**externality**). $\Delta_1 \geq 0$, with equality iff $\tilde{\theta}^* = 0$ (equivalently, $\theta^* = 0$ by (C4)). At a quadratic approximation around n^{CB} ,

$$\Delta_1 \approx \frac{1}{2} \omega (\beta \xi_N^P)^2 / \gamma,$$

where $\omega > 0$ is the curvature of welfare in fertility. ξ_N^P carries the sign of θ^* ; the squared form makes $\Delta_1 \geq 0$.

Δ_2 (**mean bias**). With $\bar{\theta} = \phi \theta_R + (1 - \phi) \theta_M$,

$$\Delta_2 \approx \frac{1}{2} \omega (n^{CB}(\bar{\theta}) - n^{CB}(\theta^*))^2 \approx \frac{1}{2} \omega \kappa^2 (\bar{\theta} - \theta^*)^2,$$

where $\kappa = \partial n^{CB} / \partial \theta > 0$ by Lemma 1.

Δ_3 (**dispersion**). Heterogeneous beliefs induce $\text{Var}(n_i) \approx \kappa^2 \text{Var}(\theta_i)$, so

$$\Delta_3 \approx \frac{1}{2} \omega \kappa^2 \text{Var}(\theta_i).$$

The expansion is taken around n^{SP} , so Δ_1 explicitly carries the externality.

Sign and magnitude properties. $\Delta_1, \Delta_2, \Delta_3 \geq 0$ by construction. Δ_2 increases in $(\bar{\theta} - \theta^*)^2$; Δ_3 increases in $\text{Var}(\theta_i)$. Public announcements affect $\bar{\theta}^{post}$ and $\text{Var}(\theta_i^{post})$ but not Δ_1 . $\square$

B.4 Proof of Theorem 4

Under W^{dyn} ,

$$W^{dyn}(\mu) = \phi_0 V_R^{post}(\mu) + (1 - \phi_0) V_M^{post}(\mu).$$

The announcement μ enters V_j^{post} only through the aggregate social credence π_t^{post} and the resulting dynamics of ϕ_t via (15), since individual types act under certainty given their own θ_j . Blackwell's theorem, applied to the aggregate-credence updating rule (12), implies that any signal that is more informative weakly increases the alignment of π_t^{post} with the truth and hence weakly increases the welfare each type assigns to the resulting future dynamics of ϕ_t . Truthful announcement $\mu = \hat{\theta}$ is most informative among signals with prior mean $\hat{\theta}$. The dynastic planner has no externality motive beyond what households themselves would weight under their own subjective valuations: W^{dyn} uses subjective beliefs, so the externality wedge does not enter. The optimal announcement is $\mu^{dyn} = \hat{\theta}$. $\square$

B.5 Proof of Theorem 5

Under W^{tot} , the announcement μ enters households' decisions only through the social posterior π_t^{post} , which through (15) shifts the next-generation type composition ϕ_{t+1} toward Romerian recruitment when $\mu > \bar{\theta}$ and toward Malthusian recruitment when $\mu < \bar{\theta}$. The shift in π_t^{post} is proportional to $\alpha = \tau_A / (\tau_A + \tau_{prior}) \in (0, 1)$. The pass-through to next-period aggregate fertility, by Lemma 1 combined with the recruitment dynamics, is approximately linear:

$$\bar{n}_{t+1}^{post}(\mu) \approx \bar{n}_t^{HB} + \alpha \kappa (\mu - \bar{\theta}),$$

where $\kappa > 0$ now summarizes the joint effect of the shift in π_t^{post} on ϕ_{t+1} and the fertility gap $n_R^* - n_M^*$. Existing households' types and fertility are unaffected by the announcement; only the composition of the next generation responds, so the announcement bites with a one-period lag relative to a model with within-head individual updating, but the functional form of the optimum survives.

The total utilitarian welfare evaluated under truth, to second order around n^{FB} , is

$$W^{tot}(\bar{n}_{t+1}^{post}) \approx W^{FB} - \frac{1}{2}\Omega(\bar{n}_{t+1}^{post} - n^{FB})^2 - \frac{1}{2}\Omega'\text{Var}(n_i^{post}).$$

Differentiating with respect to μ :

$$\frac{dW^{tot}}{d\mu} = -\Omega(\bar{n}_{t+1}^{post} - n^{FB}) \cdot \alpha\kappa + \text{terms from } \text{Var}(n_i^{post}).$$

Setting to zero and decomposing $n^{FB} - \bar{n}_t^{HB} = (n^{FB} - n^{CB}(\theta^*)) + (n^{CB}(\theta^*) - \bar{n}_t^{HB})$:

$$\mu^{tot} \approx \theta^* + \frac{\beta\xi_N^P}{\alpha\kappa\gamma}.$$

Replacing θ^* by the planner's signal $\hat{\theta}$:

$$\mu^{tot} = \hat{\theta} + \chi^{tot}, \quad \chi^{tot} = \frac{\beta\xi_N^P}{\alpha\kappa\gamma}.$$

The shade $\chi^{tot} > 0$ iff $\xi_N^P > 0$, iff $\tilde{\theta}^* > 0$, iff $\theta^* > 0$ by (C4).

Comparative statics. (i) ξ_N^P is increasing in $|\tilde{\theta}^*|$, which by (C4) is increasing in $|\theta^*|$; so $|\chi^{tot}|$ is increasing in $|\theta^*|$. (ii) Evaluating ξ_N^P at non-zero deviation of population from the reference involves a correction that, under the Taylor expansion (35), is proportional to $|\ln(N_0/\bar{N})|$. (iii) Larger $\text{Var}(\theta_{i,0})$ raises curvature Ω' (and lowers effective α), reducing χ^{tot} . (iv) Higher τ_A raises α ; the absolute welfare effect of the announcement is increasing in τ_A . $\square$

B.6 Proof of Corollary 1

Repeating the argument of Theorem 5 with W^{avg} in place of W^{tot} , the only change is that the externality wedge becomes $\beta\xi_{N,avg}^P$ instead of $\beta\xi_N^P$. By Proposition 10, $0 < \xi_{N,avg}^P < \xi_N^P$ when $\theta^* > 0$. Therefore $0 < \chi^{avg} < \chi^{tot}$. $\square$

B.7 Proof of Corollary 2

Direct from Theorem 4, Corollary 1, and Theorem 5, with $0 < \chi^{avg} < \chi^{tot}$ in a Romerian world. $\square$

B.8 Proof of Proposition 6

In the multiplicity region, the welfare derivative with respect to μ includes a term reflecting the change in the realized selection at the cutoff ϕ^{rd} . Let $g(\phi)$ denote the prior density of ϕ . The welfare gain from a marginal lowering of the cutoff in a Romerian world is

$$\frac{dW^{tot}}{d\phi^{rd}} = -g(\phi^{rd}) \cdot [W^{tot}(\bar{n}^H, \cdot) - W^{tot}(\bar{n}^L, \cdot)],$$

which is negative because the bracket is positive (high-fertility selection has higher welfare in a Romerian world by (C4)). The chain rule gives

$$\left. \frac{dW^{tot}}{d\mu} \right|_{\text{selection}} = \frac{dW^{tot}}{d\phi^{rd}} \cdot \frac{d\phi^{rd}}{d\mu},$$

where $d\phi^{rd}/d\mu < 0$ in a Romerian world (a more optimistic announcement lowers the share threshold above which the high-fertility selection is risk-dominant). The product is positive, so the selection-shifting marginal benefit is positive.

Adding this to the FOC of Theorem 5 yields an additional shading term χ^{sel} . The non-monotonicity follows from $g(\phi^{rd})$: when ϕ is far from ϕ^{rd} on either side with concentrated prior, $g(\phi^{rd})$ is small. χ^{sel} peaks when the prior places maximum mass near the cutoff. $\square$